The Making of Dominant Vehicle Currencies: Evidence from DeFi

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Abstract

We reconstruct 475 million Ethereum decentralised-exchange swaps and observe vehicles inside routes. Stablecoin use rises +25.5 percentage points (pp) from 2024 H1 to 2026 H1. Within-pair net substitution is −0.1 pp; pair entry and trading reallocation account for it. Stable-facing Uniswap v3 additions rotate across origins; prior vehicle-supply experience predicts participation in new pools. Before first stablecoin use, capital on both legs predicts +1.84 pp higher weekly adoption; conditional on support, a tenfold stablecoin/WETH weak-leg-capital ratio predicts another +1.84 pp. Relative capital predicts route allocation and persistent price leadership. Providers form funded networks, while prices revalue their capital between actions; the short-horizon revaluation does not predict monthly vehicle use.

1 Introduction

A vehicle currency is acquired for immediate resale. It becomes useful when two liquid markets through the vehicle offer a better exchange than a thin direct market. Traditional theories therefore connect vehicle use to a self-reinforcing relation between trading volume and transaction costs (Krugman, 1980; Kiyotaki and Wright, 1989; Flandreau and Jobst, 2009). The U.S. dollar appears on more than 85% of foreign-exchange transactions, far beyond the United States' share of cross-border activity (Somogyi, 2026). This demand-side account has an upstream counterpart. Market-making capacity supplies immediacy, and a shared intermediary can generate common liquidity across markets (Grossman and Miller, 1988; Coughenour and Saad, 2004). We study how liquidity providers fund a vehicle network, how that network expands into new trading relationships, and how its quotes relate to persistent vehicle use.

Aggregate vehicle dominance combines two economic margins. A continuing relationship can change its vehicle mix, while routed activity can also move toward relationships that already use another vehicle and toward new relationships formed around it. The first margin reveals substitution inside established trading relationships. The second reveals the vehicle inherited when new relationships acquire liquidity. Both can generate the same aggregate rotation, yet they imply different accounts of currency competition and persistence.

These margins are directly observable on decentralised exchanges. Standard FX records report the two legs of a customer's exchange separately, so a yen–euro exchange routed through dollars appears as yen–dollar and dollar–euro trades. Somogyi (2026) infers vehicle activity from currency-pair volumes around non-overlapping holidays. On Ethereum, pool calls within one transaction retain their execution order. We reconstruct connected routes from 475 million swaps on 9 automated market maker (AMM) deployments from November 2018–June 2026 (Table 1). We call the ordered source and destination the *endpoint pair*, shortened hereafter to *pair*. Each directed pool execution is a *leg*; the connected sequence of legs is a *route*; and an asset between the endpoints is a vehicle. The route records on-chain execution. An off-chain user instruction can begin before or end after the connected pool sequence. We consolidate ETH and WETH as one native asset and preserve each stablecoin's token identity before aggregating the stablecoin family. The headline share therefore measures tokenised-dollar intermediation, while issuer results retain competition among USDC, USDT, DAI, and other dollar-linked instruments.

Stablecoins take routed-value leadership while expanding their vehicle role mainly as the pair network changes. Between matched January–June dates in 2024 and 2026, their pooled share of two-leg routes with a native or stablecoin vehicle rises from 16.9% to 42.4%, an increase of +25.5 percentage points (pp) (Table 2, panel A). Their paired-day routed-value share reaches 76.5%, while their count share remains below one half (Appendix Table 19, panel B). Net substitution within continuing pairs contributes −0.1 pp by route count. Trading reallocation across continuing pairs contributes +8.4 pp, and pairs first observed after 2024 H1 contribute +20.1 pp before the other lifecycle margins enter the period-specific net (Table 2, panel A). Movements toward and away from stablecoins inside continuing pairs are individually substantial and almost exactly offset. The same extensive-margin pattern appears in routed value, adjacent-period comparisons, endpoint sets outside the two vehicle families, and routes with more than two legs. The vehicle used when a pair first trades also predicts its vehicle mix over the next 120 days.

Stable-facing liquidity additions rotate upstream of route use. In Uniswap v3 spoke pools that pair WETH, DAI, USDC, or USDT with an asset outside that set, the stable-facing share of addition actions rises from 8.5% in 2024 H1 to 44.2% in 2026 H1. Vehicle-side dollar additions rise from 8.1% to 41.5% (Table 4, panel A). Transaction origins active in only one comparison period contribute +28.00 pp of the action-share increase, while changes within continuing origins contribute +7.70 pp (panel A). The rotation remains after excluding each pool's first seven observed days and is concentrated in repeated participation and mature pools. At the first week a pool reaches $50,000 of reported total value locked, an origin that supplied the same vehicle elsewhere during the preceding 90 days is +7.99 pp more likely to supply it in the new pool (panel B, column 1). The comparison excludes the focal pool and every pool attached to the focal endpoint. This vehicle-specific experience suggests that liquidity networks expand through providers already familiar with the vehicle.

A complementary constant-product sample connects two-leg capital to realised vehicle use. Before a pair's first stablecoin route, measured stablecoin capital on both required legs is associated with a +1.84 pp higher weekly probability of first use. Conditional on support, a tenfold increase in stablecoin relative to WETH weak-leg capital adds +1.84 pp (Table 5, panel A, columns 1–2). Inside established pairs, a 100 basis point incumbent exact-output advantage predicts +10.13 pp higher retention after controlling for prior capital; a 10 percentage point incumbent capital-share advantage predicts another +2.77 pp (Table 6, panel A, column 2). When exact-output leadership changes hands, incumbent route share falls 29.0 pp more than during the same event's earlier movement, and prior challenger capital predicts whether the new price lead persists (panel B, columns 2 and 1). Providers establish pools and supply liquidity units. Prices, swaps, fees, transfers, and later provider actions determine the reserve stock and its dollar value; routers compare size-specific quotes.

These findings give the classical volume–cost mechanism a liquidity-network foundation. The pair decomposition shows that dominance can rotate through network formation even when net substitution inside continuing relationships is nearly zero. Each design measures a different link: Uniswap v3 identifies provider specialisation, Uniswap v2 and SushiSwap v2 measure full-range capital, and those venues together with Uniswap v3 measure executable prices. These links connect funded networks to route use and persistent price leadership. Funded links and liquidity units change with provider actions, fees, and transfers; market prices can revalue standing positions between external provider actions. At one- to seven-day horizons, ETH declines raise stablecoin-relative dollar capital in fixed constant-product pools, while the relative $\sqrt{k}$ estimates are small and imprecise (Appendix Table 36, columns 1–3). This comparison says little about gross provider activity or within-range management. Monthly coefficients for route-specific weak-leg capital, exact output, and vehicle use are near zero (Appendix Table 35, panel B, row 1). These results separate short-run mark-to-market revaluation from the formation and placement margins associated with the longer-run rotation. Demand for dollar-linked assets, issuer-supported convertibility, provider expertise, expected fees, and inventory risk can all influence that allocation. Our data measure pool formation and route execution; provider motives remain an economic interpretation. The same distinction appears in traditional accounts in which demand for dollar funding, trade finance, and safe assets reinforces the dollar's international role (Gopinath and Stein, 2021; Chahrour and Valchev, 2022).

The connected route measure complements historical and holiday-based evidence on vehicle use and extends AMM research from isolated pools to sequential exchange through competing vehicles (Lehar and Parlour, 2025; Xi and Moallemi, 2026). Section 2 defines the route measure, institutional setting, and data coverage. Section 3 establishes the rotation, its pair margins, and post-entry persistence. Section 4 studies liquidity provision and two-leg capital formation. Section 5 compares capital, exact prices, and route allocation. Section 6 develops the monetary and market-structure implications.

2 Vehicle currencies on decentralised exchanges

2.1 Routes and notation

Decentralised exchanges distribute liquidity across smart-contract pools that routers can combine within one transaction. A trader may use one pool, divide an order across several pools, or trade through an intermediary asset. Because the pool calls share a transaction, the intermediary sequence is observable.[1] An intermediary route supplies connectivity when a direct pool is absent; when direct and indirect pools coexist, the router compares their fees, price impact, gas, and venue access. Lehar and Parlour (2025) describe the constant-product design and the economics of liquidity supply in these markets.

Blockchain logs record each pool call in execution order. We recover a route by following the directed token flows within a transaction. Calls with a common source and destination form a split order; a token received in one call and spent in a later call joins the calls into a sequential route. The same rule separates unrelated conversions bundled in one transaction.

[1] Ethereum records a transaction's ordered logs in its receipt; see the official Ethereum JSON-RPC documentation.

For a reconstructed sequence $A \to B \to C$, (A, C) is the pair, $A \to B$ and $B \to C$ are its legs, and the full sequence is the route. All three objects are directed by token flow. We reserve *route* for an observed connected execution and *path* for a feasible ordered sequence of legs. Each leg executes in a liquidity pool, and some pairs are connected only through a vehicle. The observed route endpoints belong to the connected pool component and may differ from the endpoints of a broader user instruction.

A transaction on January 10, 2026 illustrates both the information in the logs and its limit. A public explorer describes a 460,000 PYUSD–USDe swap executed through 0x.[2] The user's instruction begins with PYUSD, but the connected pool sequence begins when 459,929.81 USDC enters Fluid. Fluid returns 460,331.05 USDT, which then enters Uniswap v4 and produces 460,104.28 USDe. The observed pair is therefore USDC $\to$ USDe; its legs are USDC $\to$ USDT and USDT $\to$ USDe, and USDT is the vehicle. The earlier PYUSD $\to$ USDC conversion occurs outside the connected pool sequence. The role of USDC in that earlier conversion remains unobserved: it may be another vehicle or executor inventory. We restrict reconstruction to the connected pool sequence and leave nearby transfers unassigned. This choice may understate the intermediary role of assets such as USDC when a broader instruction is funded outside the observed pool sequence.

The sequence identifies intermediary use directly. Reconstructing each pool immediately before execution also lets us compose exact-input quotes along the observed route and feasible alternative paths at the same notional. Appendix Equation (6) gives the route composition, Appendix B gives each pool-family quote, and Section 5 applies them to stablecoin and WETH paths.

This intermediary role is distinct from invoicing and funding (Gopinath et al., 2020; Gopinath and Stein, 2021; Mukhin, 2022). It makes the classical volume–cost mechanism observable at transaction level: use expands trading on the two supporting legs, while deeper markets can lower the cost of using the same vehicle again (Krugman, 1980; Kiyotaki and

[2] Transaction `0xbda1d07f...9bf67ad9`.

Wright, 1989; Dowd and Greenaway, 1993). Liquidity placement, venue access, and available paths provide the corresponding coordination forces on decentralised exchanges.

2.2 Measuring vehicle dominance

An exact two-leg route has one pool trade on each side of a unique intermediary and therefore one mutually exclusive vehicle choice. A longer route can use several intermediaries. We retain both objects: the native-versus-stable decomposition uses exact two-leg routes, while the wider time series counts every intermediary position and also records whether an asset appears anywhere inside the route.

Routes are reconstructed at the token level before tokens are assigned to economic groups. The native group consolidates ETH with its one-for-one wrapped representation, WETH; liquid-staking tokens remain separate. Dollar-pegged tokens also remain separate instruments because issuer, backing, redemption, risk, and leg-level liquidity differ (Gorton and Zhang, 2023; Catalini et al., 2022; Lyons and Viswanath-Natraj, 2023). The stable group then aggregates those token-level instruments for the family comparison, while issuer results retain USDC, USDT, DAI, and other stablecoins separately.

On each day, the count measure is stablecoin routes divided by all exact two-leg routes whose intermediary is native or stable. The value measure weights the same choices by the mean dollar value at the observed source and destination. It retains only routes for which source, intermediary, and destination values agree within 20%; every intermediary's incoming and outgoing values must also lie within that band. Appendix Equation (9) gives the exact shares. Levels are percentages, and absolute changes are in percentage pointpercentage points (pppp). Staked-native, imported, and other intermediaries remain in Figure 1 and Appendix Table 26; they stay outside the native-versus-stable denominator.

Throughout, *vehicle dominance* denotes the intensity of intermediary use. We reserve *leadership* for first rank under a named count or value measure; graph centrality is a separate network statistic. An excess-use ratio divides an asset's intermediary share by its endpoint share and therefore distinguishes its role in the middle of routes from ordinary source-or-destination demand.

2.3 The route panel

The route panel covers Curve, Uniswap v1 through v4, Balancer, SushiSwap v2 and v3, and Fluid on Ethereum. It spans 2,798 dates from November 2018–June 2026 and includes constant-product, concentrated-liquidity, stable-swap, weighted-pool, singleton, and integrated-liquidity designs. Table 1 summarizes the resulting sample. Indexed blockchain records supply eight exchange series, Dune supplies Fluid, and direct Ethereum queries supply block headers, ordered logs and receipts, token precision, registries, and state checks. Appendix Table 16 reports coverage relative to Ethereum exchange volume.

The source contains 475,071,108 pool swaps and yields 474,388,425 usable directed legs after event direction and token units are standardised (Table 1). Factory records usually identify pool tokens; for Uniswap v1, the immutable exchange registry supplies the token address and matched ETH flows order the two forced legs. Appendix Table 9 contrasts Uniswap v1's mandatory ETH links with the arbitrary ERC-20 pools permitted by Uniswap v2. Within every transaction, the directed-flow rule distinguishes single-pool trades, split orders, sequential routes, and unrelated calls. Appendix A gives the reconstruction details.

Transactions with more than one disconnected pool-event group pose a separate measurement limit because the observed calls leave the broader user instruction unknown (Xi and Moallemi, 2026). Our route sample retains transactions with a single connected group. Appendix Table 7 treats each internally valid component as a separate route and finds a slightly larger stablecoin rotation.

Every non-self-returning sequence retained by the connected-transaction rule enters the count measure. Dollar-weighted results add the 20% value-agreement rule, which selects routes with a coherent dollar notional. Appendix F reports the separate round-trip exclusion, and each value exhibit reports its sample coverage.

Transactions and pool-state changes are publicly observable (Schär, 2021). The reconstruction identifies each executed route and its calling contract; the economic unit remains the executed route because an executor address can serve several frontends or quoting systems (Xi and Moallemi, 2026).

Full-day reconciliation against Ethereum logs covers 206 venue-days across 74 dates. The Uniswap v2, SushiSwap v2, and Uniswap v3 comparisons span 13.9 million indexed swaps. Swap-event precision is 100.0000%, 100.0000%, and 99.9998%, respectively; swap-event recall is 99.9995%, 100.0000%, and 99.9335% (Appendix Table 8, panel A). Reconstructing the corrected days changes stablecoin share by −0.008 pp by route count and −0.016 pp by routed value (panel C). Every component of the sampled 2024–2026 decomposition moves by at most 0.001 pp (panel D). A separate receipt sample confirms the token direction and reported amounts of all 245 sampled Fluid legs; the Wilson 95% lower bound for label precision is 98.5% (panel E). The checks validate event coverage and route labels on their sampled observations. Appendix A gives the remaining construction checks, and Appendix B validates the pretrade pricing models.

Table 1: The route panel

	Route panel
Dates in the full route panel	2,798
Missing source-days	0
Raw swap rows	475,071,108
Usable directed route legs	474,388,425
Routes per day, median across eligible days	153,586
Multi-leg routes per day, median across eligible days	30,064
Multi-leg share of routes, median across eligible days	19.4%
Round trips, median share of multi-leg routes by count	12.8%
Round trips, median share of multi-leg routes by value	21.0%

Note: The full 9-deployment panel contains UTC dates from November 2018–June 2026. Uniswap v1 enters these totals through the exact exchange-to-token registry; its protocol-architecture comparison is venue-specific. Daily route statistics and round-trip shares use all 2,449 eligible dates from 15 May 2019 through 30 June 2026. A multi-leg route traverses more than one pool; a round trip ends in the token with which it began. Round-trip shares are formed among multi-leg routes.

3 The rotation in vehicle dominance

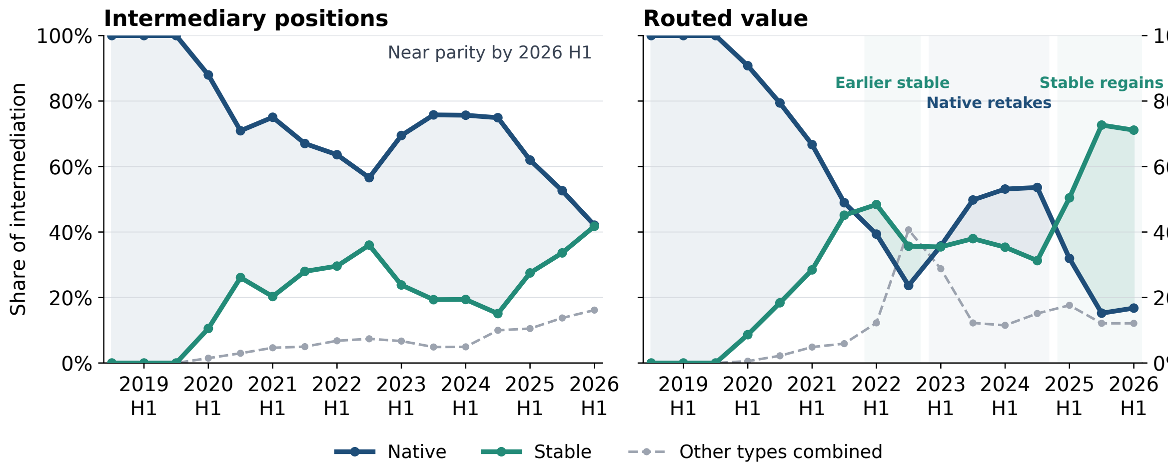
3.1 The aggregate rotation

Stablecoins carry a much larger share of vehicle activity in 2026 than in 2024. The comparison uses the same January–June dates in both years because the 2026 panel ends on June 30. Giving each matched day equal weight, stablecoins' share among native and stable intermediaries rises from 16.9% to 42.1% by route count and from 32.7% to 76.5% by routed value. Appendix Table 19 reports the paired-day estimates and their sampling uncertainty. The decomposition below pools route activity so its components allocate the aggregate count or value exactly.

The half-year series in Figure 1 reverses direction. Native assets regain share in 2023 and 2024; stablecoins recover the routed-value lead in 2025. Their count share remains below one half in 2026 H1. WETH also retains first rank in count-weighted network centrality and in value-weighted centrality after stablecoin-core legs are removed (Appendix Tables 26 and 27). Stablecoins therefore lead by routed value on the full route perimeter, while WETH remains the leading connector under those count and non-core graph measures.

The increase also exceeds ordinary endpoint demand for USDT. Its excess-use ratio rises from 1.06 to 1.22 by count and from 0.59 to 1.40 by supported value. Across paired January–June dates, USDT's intermediary share minus its endpoint share moves from −7.13 pp to +7.95 pp (Appendix Table 20). The issuer's expanding endpoint role therefore accounts for only part of its vehicle transition.

Figure 1: Intermediary activity rotates between native assets and stablecoins



Note: The sample contains complete non-self-returning routes. Each route contributes once for every intermediary it uses, so routes with more than two legs can contribute several observations. Shares are formed across all intermediary types; Appendix Table 19 uses the narrower native-or-stable denominator. Every point covers six calendar months; the final point is 2026 H1. Routed value requires source, intermediary, and destination dollar amounts to agree within 20%.

3.2 Pair entry and activity reallocation

Aggregate dominance can change inside continuing pairs or through the distribution of activity across pairs. Historical vehicle-currency evidence likewise links a currency's aggregate role to the trading relationships it connects (Flandreau and Jobst, 2009). Let p denote a pair and $y \in \{2024, 2026\}$. A continuing pair carries native-or-stable routes in both matched periods; a year-specific pair carries them in only one. First split total activity between these two sets. Within continuing pairs, stablecoin share can change because a given pair changes its vehicle mix or because trading shifts toward pairs with a different vehicle mix. The remaining change comes from the aggregate weight on continuing pairs and the stablecoin share of pairs observed in only one period.

In each product, the exact midpoint decomposition holds one factor at its two-period average while the other changes. Its four terms add to the pooled rotation: vehicle switching within continuing pairs, activity shifting across those pairs, the aggregate weight on continuing pairs, and the stablecoin share of pairs present in only one period. We also compare the same pair and month-day, separately for routes whose legs use one exchange or several, with pair-by-calendar-cell effects. Appendix G gives the algebra and matched specification.

Table 2 applies both comparisons to route counts and supported value.

Table 2: Decomposition of the stablecoin rotation

Component or estimate	Estimate [pp]	Obs.
Panel A. Route-count share: continuing and year-specific pairs		
Net stablecoin-share change within continuing pairs	−0.1	
Pairs moving toward stablecoins (1,569)	+1.3	
Pairs moving toward native assets (1,487)	−1.4	
Vehicle activity shifting across continuing pairs	+8.4	
Weight of continuing versus year-specific pairs	−0.5	
Net contribution of period-specific vehicle activity	+17.79	
Pairs first observed after 2024 H1	+20.09	
Pairs reactivated after absence in 2024 H1	+0.20	
Vehicle-role turnover in continuing pairs	−0.77	
Pairs exiting before 2026 H1	−1.73	
Total route-count change	+25.5	
Panel B. Dollar-weighted share: continuing and year-specific pairs		
Net stablecoin-share change within continuing pairs	0.0	
Pairs moving toward stablecoins (1,505)	+2.3	
Pairs moving toward native assets (1,445)	−2.4	
Vehicle activity shifting across continuing pairs	+26.2	
Weight of continuing versus year-specific pairs	−2.5	
Net contribution of period-specific vehicle activity	+19.16	
Pairs first observed after 2024 H1	+21.94	
Pairs reactivated after absence in 2024 H1	+0.04	
Vehicle-role turnover in continuing pairs	−1.72	
Pairs exiting before 2026 H1	−1.10	
Total change in dollar-weighted share	+42.8	
Panel C. Matched pair estimates		
All two-leg routes, count share	+0.23 (0.77)	188,344
20% agreement sample, count share	+0.32 (0.75)	182,734
20% agreement sample, dollar-weighted share	−1.35 (2.19)	182,734

Note: Panels A and B compare pooled activity over matched January–June dates in 2024 and 2026 and report the exact midpoint decomposition in Appendix Equation (10). Their first indented rows split the within-pair component into positive and negative contributions. Their lifecycle rows split the net contribution of period-specific vehicle activity using each pair's first and last observation in the full panel: first-observed entry, reactivation after absence in 2024 H1, vehicle-role turnover among pairs active in both periods, and exit before 2026 H1. These lifecycle rows add to the net term using unrounded values. Panel B weights routes by dollar value and applies the 20% agreement rule. Panel C estimates Appendix Equation (11); standard errors use two-way clustering by pair and date. Appendix Table 23 reports an alternative factorisation of the route-count change.

Pooling route counts gives an increase from 16.9% to 42.4%, or +25.5 pp, close to the equal-day change in Appendix Table 19. The within-pair component is −0.1 pp by route count and 0.0 pp by routed value. These net estimates combine +1.3 pp from pairs moving toward stablecoins and −1.4 pp from pairs moving toward the native asset. Activity shifting across continuing pairs contributes +8.4 pp by count and +26.2 pp by value. The changing aggregate weight on continuing pairs contributes −0.5 pp and −2.5 pp (Table 2, panels A and B).

The period-specific contribution is also a net. Pairs first observed after 2024 H1 contribute +20.1 pp by route count and +21.9 pp by routed value. Reactivation, vehicle-role turnover, and exit together contribute −2.3 pp and −2.8 pp, leaving the period-specific terms at +17.8 pp and +19.2 pp. First-observed pair entry therefore supplies 78.9% of the total route-count rotation and 51.2% of the dollar-weighted rotation (Table 2, panels A and B). The matched specification reaches the same conclusion on the more restrictive sample: stablecoin share changes by +0.2 pp by count and −1.4 pp by value (panel C). A large aggregate rotation therefore coexists with −0.1 pp net switching after holding constant the pair, calendar date, and realised single- or cross-venue route class.

Pair-period materiality floors preserve the same allocation. Requiring at least five or ten routes, or at least $5,000 or $50,000 of supported value, leaves the within-pair term close to zero; reweighting and period-specific pairs continue to supply essentially the full rotation (Appendix Table 21). This comparison removes pair-period cells supported only by isolated or economically small activity.

The near-zero within-pair component also survives venue exclusions. Its route-count estimate ranges from −1.06 to +1.03 percentage points when we remove one venue family at a time. Uniswap v4 contributes substantially to the count increase, while routed value still rises by 31.34 percentage points after removing v4 and by 21.74 points on Uniswap v2/v3 and SushiSwap v2 alone (Appendix Table 22).

The pattern extends through time and across endpoint sets. In every adjacent January–

June comparison from 2019 through 2026, the within-pair term lies between −0.4 pp and +1.0 pp even as the aggregate moves from −6.4 pp in 2022–2023 to +17.4 pp in 2025–2026 (Appendix Table 24, panel A). Removing every pair with WETH or a stablecoin at either endpoint still gives an increase from 1.1% to 9.0% by count and from 0.9% to 39.1% by supported value (panel B). Appendix Table 25, panel B, then allocates the headline change across endpoint directions and shows that USDT supplies +13.7 pp of the routed-value increase between two stablecoin endpoints. Pair composition therefore accounts for stablecoin gains and reversals even when endpoint identity leaves both broad vehicle families available.

Across matched route cells, adjacent periods, and endpoint sets, activity moving among pairs accounts for most of the aggregate change. Appendix G gives the decomposition algebra and reports the adjacent-period, endpoint, issuer, and network results.

3.3 Persistence in observed use after pair entry

The vehicle used when a pair enters strongly predicts its subsequent use. For pair p, let E_p be the stablecoin share on its entry day. For each of the disjoint windows $h \in \{1\text{–}30, 31\text{–}120\}$, let $R_{p,h}$ indicate that the pair carries another native- or stablecoin-intermediated route, and let $S_{p,h}$ be the stablecoin share when $R_{p,h} = 1$. We estimate

$$\begin{aligned} S_{p,h} &= \alpha_h + \beta_h E_p + X_p'\delta_h + \varepsilon_{p,h}, \quad R_{p,h} = 1, \\ R_{p,h} &= a_h + b_h E_p + X_p' c_h + u_{p,h}, \quad \text{all entrant pairs.} \end{aligned} \tag{1}$$

The second line is a linear probability model. The controls X_p contain the cohort year, a stablecoin-endpoint indicator, log entry-day route count, direct-route share, and complex-route share. The sample uses pairs entering by March 2 in 2024 or 2026, giving every pair 120 complete calendar days of follow-up, and keeps pairs whose endpoints lie outside the native group.

Table 3: Persistence in observed vehicle use after pair entry

Panel A. Stablecoin route share among pairs that trade again

	Days 1–30			Days 31–120		
	(1)	(2)	(3)	(4)	(5)	(6)
Entry stablecoin share [effect pp per 10 pp]	+9.00***	+8.92***	+8.55***	+8.59***	+8.40***	+9.00***
	(0.14)	(0.15)	(0.72)	(0.18)	(0.18)	(0.50)
Controls	No	Yes	Yes	No	Yes	Yes
Route-activity weights	No	No	Yes	No	No	Yes
Observations (pairs)	30,547	30,547	30,547	19,405	19,405	19,405
Entry-date clusters	123	123	123	123	123	123
Mean stablecoin share [pp]	2.70	2.70	4.78	3.83	3.83	7.18
R^2	0.842	0.843	0.800	0.787	0.791	0.830

Panel B. Subsequent-trading incidence among all entrants

	Days 1–30	Days 31–120
Entry stablecoin share [effect pp per 10 pp]	+0.30***	+0.98***
	(0.08)	(0.08)
Controls	Yes	Yes
Observations (pairs)	157,262	157,262
Entry-date clusters	123	123
Mean retrading rate [pp]	19.42	12.34
R^2	0.068	0.033

Note: Panel A estimates the first line of Equation (1) among pairs that carry another native- or stablecoin-intermediated route in the stated window. Panel B estimates the second line on all entrant pairs. The entry day is excluded from both follow-up windows. Controls are the cohort year, a stablecoin-endpoint indicator, log entry-day route count, direct-route share, and complex-route share. Columns 3 and 6 weight pairs by route activity in the respective follow-up window; all other columns give each pair equal weight. Coefficients report the outcome change in percentage points (pp) associated with a 10 pp higher entry-day stablecoin share. Standard errors in parentheses use CR1 clustering by entry date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

We separate later vehicle use from pair survival. Of the 157,262 entrants, 19.4% trade again during days 1–30 and 12.3% do so during days 31–120 (Table 3, panel B). Among pairs that trade again, a 10 pp higher entry-day stablecoin share predicts +8.92 pp higher stablecoin share during days 1–30 and +8.40 pp during days 31–120 (panel A, columns 2 and 5). The route-activity-weighted estimates are +8.55 pp and +9.00 pp, respectively (columns 3 and 6). Every follow-up outcome begins after the entry day, so these estimates capture continued vehicle use across subsequent routes.

Entry-day stablecoin share also predicts a modestly higher probability of subsequent trading: a 10 pp increase corresponds to +0.30 pp during days 1–30 and +0.98 pp during days 31–120 (Table 3, panel B). Vehicle identity therefore persists strongly among pairs active in each window, alongside a smaller survival margin. This is persistence in realised use, including pairs for which the rival vehicle may remain unavailable. Only 0.5% of material entrants reach a sampled exact contest between both vehicle families, with a median lag of 115.5 days (Appendix Table 37, panel A). Section 5 studies price leadership when such a contest is observed. The estimates remain similar among pairs with more entry-day trades (Appendix Table 30). Relative prices, depth, and venue access can shape the vehicle at entry and remain relevant for later trading, as in models of currency competition with network effects (Dowd and Greenaway, 1993).

4 Liquidity provision and network formation

Routers can choose only among paths whose two legs have already received capital. Liquidity providers choose which links to fund, and automated routers compare the resulting size-specific quotes when selecting a route. Liquidity supply is voluntary and responds to expected trading conditions (Li et al., 2021). In concentrated-liquidity pools, providers also choose the price range in which their capital is active (Lehar and Parlour, 2025). We study two complementary samples. Uniswap v3 additions reveal the participation and prior experience of transaction origins that supply stablecoin- and WETH-facing pools. Full-range Uniswap v2 and SushiSwap v2 reserves then date the appearance of two-leg stablecoin capital before the first observed stablecoin route. The samples connect provider participation, capital placement, and route use without tracking the same position across pool designs.

4.1 Stable-facing liquidity additions

Uniswap v3 records each addition to a liquidity position and the transaction origin that initiates it. We call a pool a *spoke* when it pairs exactly one of WETH, DAI, USDC, or USDT with an asset outside that set; a *stablecoin core* pool pairs two of DAI, USDC, and USDT. The stable-facing share of spoke-pool addition actions rises from 8.5% in 2024 H1 to 44.2% in 2026 H1. The share of vehicle-side dollar additions follows a similar rotation, from 8.1% to 41.5% (Table 4, panel A). Thus the change in routed activity coincides with a large upstream shift in gross liquidity additions.

We decompose this change across transaction origins using the same midpoint identity as the pair decomposition. Period-specific origins contribute +28.00 pp of the +35.71 pp increase in the stable-facing action share, while changes within continuing origins contribute +7.70 pp. The corresponding contributions for dollar additions are +17.67 pp and +15.86 pp (Table 4, panel A). A transaction origin measures participation in the liquidity addition; beneficial ownership and coordinated control remain unobserved.

Pool maturity separates this rotation from short-lived launches. Pools older than 90 days supply 79.5% of the increase in stable-facing addition actions (Appendix Table 28, panel A). After excluding each pool's first seven observed days, the stable-facing action share rises from 9.0% to 47.4% (panel B). Among origins present only in 2026 H1, one-day, one-pool participation supplies 3.58% of stable-facing actions and 0.49% of vehicle-side dollar additions (panel C). Mature pools and repeated participation therefore carry most of the rotation in addition activity.

The formation of a material spoke pool also draws disproportionately on origins with experience supplying the same vehicle elsewhere. For every endpoint–vehicle pool, we date the first week in which reported total value locked reaches \$50,000. We then compare the vehicles supplied by each transaction origin during that week and measure its supply during the preceding 90 days, excluding the focal pool and all pools attached to the focal endpoint. For formation event e, origin o, and vehicle v, let Y_{eov} equal one when v is the vehicle in the newly material pool. Let $H_{ov,e-1}$ denote the origin's prior same-vehicle supply elsewhere in the network, measured either as an indicator or as log endpoint breadth. We estimate

$$Y_{eov} = \beta H_{ov,e-1} + \alpha_{eo} + \lambda_{v,q(e)} + \varepsilon_{eov}, \tag{2}$$

where α_{eo} are formation-event-by-origin effects and $\lambda_{v,q(e)}$ are vehicle-by-calendar-quarter effects. Prior supply of the same vehicle predicts a +7.99 pp higher probability that the origin supplies that vehicle in the new pool (Table 4, panel B, column 1). A one-unit increase in log prior endpoint breadth predicts +3.07 pp higher probability (column 2). Both estimates remain statistically significant after Holm adjustment across the four reported models. This specialisation is consistent with existing vehicle networks supplying new spokes. Common market makers can similarly link liquidity conditions across securities (Coughenour and Saad, 2004).

Within stablecoins, prior supply to a same-token core pool has a +7.74 pp coefficient (Table 4, panel B, column 3), and prior same-token core-pool breadth has a +15.98 pp coefficient (column 4). The first estimate is imprecise after multiplicity adjustment, while the breadth estimate is significant at the 10% level. Cross-vehicle specialisation is precisely estimated; the stablecoin-core estimates are weaker and suggestive of core dollar-token liquidity extending into spokes. We observe the pools that receive liquidity; a provider choice model would additionally require each participant's full opportunity set.

Table 4: Liquidity additions and vehicle-specific pool formation

Panel A. Stable-facing liquidity additions

	Stable-facing share [%]		Change [pp]			
Supply measure	2024 H1	2026 H1	Total	Within continuing origins	Across continuing origins	Period-specific origins
Liquidity-addition actions	8.5	44.2	+35.71	+7.70	+0.01	+28.00
Vehicle-side USD additions	8.1	41.5	+33.48	+15.86	-0.05	+17.67
Active transaction origins, stable-facing	3,841	8,721				
Active transaction origins, WETH-facing	27,112	8,696				

Panel B. Prior vehicle experience and pool formation

	(1)	(2)	(3)	(4)
Compared vehicles	WETH, DAI, USDC, USDT		DAI, USDC, USDT	
Prior same-vehicle supply outside focal endpoint	+7.99*** (0.89)			
Log prior same-vehicle endpoint breadth, $\ln(1+n)$		+3.07*** (0.92)		
Prior same-stablecoin core supply			+7.74 (5.90)	
Log prior same-stablecoin core-pool breadth, $\ln(1+n)$				+15.98* (7.30)
Dependent-variable mean [%]	25.0	25.0	33.3	33.3
Observations	258,048	258,048	24,396	24,396
Pool clusters	6,197	6,197	1,807	1,807
Transaction-origin clusters	43,473	43,473	6,312	6,312
Pool formation × transaction-origin effects	Yes	Yes	Yes	Yes
Vehicle × calendar-quarter effects	Yes	Yes	Yes	Yes
Two-way clustered standard errors	Pool, origin	Pool, origin	Pool, origin	Pool, origin

Note: Panel A decomposes the 2024 H1–2026 H1 change in the stable-facing share of positive Uniswap v3 liquidity additions on spoke pools with exactly one of WETH, DAI, USDC, or USDT. Stable-facing combines DAI, USDC, and USDT. Vehicle-side dollar additions value the amount of the vehicle token; observations with missing prices or implausibly large values comprise less than 1% of additions in every period–vehicle cell. Panel B estimates Equation (2) in the first week a pool reaches $50,000 of reported total value locked. For each transaction origin supplying that pool, the dependent variable equals one for the pool's observed vehicle and zero for the other vehicles. Prior supply occurs during the preceding 90 days and excludes the focal pool and every pool paired with the focal endpoint. Columns 1–2 compare WETH, DAI, USDC, and USDT; columns 3–4 compare DAI, USDC, and USDT during stable-facing pool formation. A stablecoin core pool pairs two of DAI, USDC, and USDT. Coefficients and standard errors are in percentage points. Models absorb pool-formation-by-transaction-origin and vehicle-by-calendar-quarter fixed effects. Standard errors are two-way clustered by pool and transaction origin. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The stars use Holm-adjusted p-values across the four models. Transaction origin measures participation; beneficial ownership remains unobserved.

4.2 ETH prices and deposited capital

Provider choices establish the pools and their liquidity units. Market prices and reserve adjustment can then change dollar capital before another provider action. We separate these margins in Uniswap v2 and SushiSwap v2 by selecting, for each nonstable endpoint and date, material stablecoin-facing and WETH-facing pools and holding that pool set fixed for one, three, or seven days. For each vehicle family, a symmetric decomposition assigns the log change in dollar capital, V, to changes in square-root reserves, $\sqrt{k}$, and unit value, $V/\sqrt{k}$. Square-root reserves can change through provider actions, retained swap fees, and token transfers. Unit value combines token-price revaluation with reserve adjustment. Cross-rate arbitrage changes reserve ratios; it need not equalise capital across pools (Daian et al., 2020).

A 0.10-log-point fall in ETH's dollar price, about 9.5%, corresponds to about 4–5% more stablecoin-relative dollar capital over the three horizons (Appendix Table 36, column 1). The $\sqrt{k}$ component ranges from -1.8% to -0.8% and is imprecise after Holm adjustment (column 2). Unit value contributes +6.14%, +5.48%, and +5.26% after one, three, and seven days (column 3). The three- and seven-day estimates are numerically close to the conditional +5.00% constant-product reference. That reference is an accounting implication under its stated price and reserve assumptions; cross-price convergence is a separate transaction-level object. For 96.7% of primary intervals, WETH-side capital equals twice the WETH reserve value because the endpoint lacks an anchored price; the WETH price therefore enters both predictor and outcome. This is an accounting decomposition. Decoded external flows appear in Appendix Table 35, panel A. Daily closing states supply the comparison, and only pools with a surviving validated state enter each horizon.

The monthly route sample separately relates trailing ETH returns to prior-day V2 and SushiSwap v2 weak-leg capital, exact output for the observed trade size, and realised vehicle choice. Its stable-relative capital coefficient is -0.0005, and the exact-output and route-use coefficients are +0.15 bp and +0.07 pp (Appendix Table 35, panel B, first row). Pool membership and the binding leg can change between monthly observations; all three coefficients are near zero.

4.3 Full-range constant-product capital before route use

The classical vehicle-currency mechanism links trading volume to lower transaction costs (Krugman, 1980). Here an indirect route needs two legs: one connecting the source to the vehicle and another connecting the vehicle to the destination. We measure deposited capital as validated full-range reserve value in Uniswap v2 and SushiSwap v2 pools. The weaker leg determines this two-leg capital measure. Dollar capital changes with reserve quantity and valuation; square-root reserves can change through provider actions, retained fees, and token transfers. Deposited capital is an equilibrium stock on those constant-product venues; size-specific pretrade quotes provide the closer measure of executable liquidity across the broader venue set. Concentrated-liquidity providers can also change executable depth by moving capital along the price curve (Adams et al., 2021; Lehar and Parlour, 2025).

We follow each pair by calendar week before its first observed route through DAI, USDC, or USDT. Neither endpoint belongs to the native or stablecoin vehicle family, every pair has recent WETH-mediated activity, and pairs that never adopt remain in the weekly sample. Let A_{pt} equal one when pair p, with endpoints i and j, first uses a stablecoin during week t. Let $K_{av,t-1} = K_{va,t-1}$ denote full-range capital at the start of the week on the leg joining endpoint a to vehicle v. For $\mathcal{S} = \{\mathrm{DAI}, \mathrm{USDC}, \mathrm{USDT}\}$, Equation (3) defines the vehicle capital measures and adoption model:

$$\begin{aligned} B^S_{pt} &= \max_{v \in \mathcal{S}} \min\{K_{iv,t-1}, K_{jv,t-1}\}, \qquad B^W_{pt} = \min\{K_{iW,t-1}, K_{jW,t-1}\}, \\ A_{pt} &= \beta Z_{pt} + \alpha_p + \lambda_t + g(a_{pt}) + \gamma' X_{pt} + \varepsilon_{pt}. \end{aligned} \tag{3}$$

Here W denotes WETH, while B^S_{pt} and B^W_{pt} are stablecoin and WETH weak-leg capital. Depending on the column, Z_{pt} is any positive stablecoin support, the stablecoin-to-WETH weak-leg capital ratio among positive-support weeks, or stablecoin's share of joint weak-leg capital. Pair and week effects absorb persistent pair conditions and market-wide changes; $g(a_{pt})$ controls flexibly for pair age, and X_{pt} contains WETH weak-leg capital and recent WETH-route activity.

Any measured stablecoin capital on both legs at the start of the week predicts +1.84 pp higher probability of first stablecoin use (Table 5, panel A, column 1). Among repeated positive-support weeks, a tenfold increase in the stablecoin-to-WETH weak-leg capital ratio predicts +1.84 pp higher adoption (column 2). The stricter recent-activity sample gives +3.20 pp and +3.55 pp, respectively (panel B, columns 1–2). The presence of measured capital on both legs and its scale relative to WETH therefore provide separate extensive and intensive margins before first use.

A 10 pp larger preweek stablecoin capital share predicts +3.60 pp higher adoption (Table 5, panel A, column 3). The next-week association is +5.01 pp (column 4); when both enter together, the coefficients are +2.94 pp and +4.58 pp (column 5). Measured stablecoin capital can precede first use, while next-week capital also co-moves with adoption. The next-week measure combines provider decisions, valuation, reserve adjustment, and any response to use, so the joint timing is consistent with feedback around formation.

A complementary threshold comparison dates the first week with at least $10,000 on both stablecoin legs and follows subsequent route use and capital shares (Appendix Table 31). Persistent-support and first-use definitions appear in Appendix Table 32 and Figure 3. On the first-use day, 92.5% of stablecoin weak-leg capital lies in pools active one week earlier; continuing-pool capital also rises more on the stablecoin side than on the matched WETH side (Appendix Table 33). These timing patterns are consistent with providers anticipating route demand, traders waiting for sufficient capital on both legs, reserve revaluation, or these forces operating together (Li et al., 2021).

Table 5: Two-leg constant-product capital before first stablecoin route use

	(1) Any support	(2) Capital ratio	(3) Preweek share	(4) Next-week share	(5) Joint timing
Panel A. Prior 28 days: at least 10 WETH routes on three days					
Any measured V2 stable bridge capital before the week [0/1]	+1.84*** (0.32)				
Stable/WETH weak-leg capital ratio, positive-support weeks [10×]		+1.84*** (0.51)			
Preweek stable share of joint weak-leg capital [10 pp]			+3.60*** (1.30)		+2.94** (1.21)
Next-week stable share of joint weak-leg capital [10 pp]				+5.01*** (1.39)	+4.58*** (1.32)
Pair-weeks	325,448	8,990	325,448	318,522	318,522
Ordered endpoint pairs	57,344	1,949	57,344	56,074	56,074
First stable-route adoptions	697	270	697	692	692
Panel B. Prior 28 days: at least 50 WETH routes on five days					
Any measured V2 stable bridge capital before the week [0/1]	+3.20*** (0.92)				
Stable/WETH weak-leg capital ratio, positive-support weeks [10×]		+3.55*** (1.25)			
Preweek stable share of joint weak-leg capital [10 pp]			+8.26 (5.30)		+7.92 (5.12)
Next-week stable share of joint weak-leg capital [10 pp]				+3.45 (3.27)	+1.82 (3.19)
Pair-weeks	25,777	1,370	25,777	25,302	25,302
Ordered endpoint pairs	5,020	319	5,020	4,931	4,931
First stable-route adoptions	140	70	140	137	137

Note: The unit is an ordered endpoint-pair week before the pair's first observed DAI-, USDC-, or USDT-mediated route. Neither endpoint is WETH or one of those stablecoins. Every pair-week has recent WETH-mediated activity, and stablecoin weak-leg capital may equal zero. Capital is prior-calendar full-range reserve value in Uniswap v2 and SushiSwap v2 at the start of the week; each vehicle's two-leg measure is its weaker leg, and stablecoin capital is the largest value across DAI, USDC, and USDT. The outcome is first stablecoin-mediated route use during the week. Column 1 compares positive measured v2 stablecoin support with zero measured v2 support. Column 2 retains pairs observed for at least two positive-support weeks and reports a ln(10) increase in the stablecoin-to-WETH log-capital advantage. Columns 3–5 use stablecoin's share of joint stablecoin and WETH weak-leg capital. Linear probability models absorb pair and calendar-week fixed effects, include pair-age bins, log WETH weak-leg capital, and prior WETH-route activity, weight pair-weeks equally, and cluster standard errors by pair and week. Columns 4 and 5 measure next-week capital, which can include adjustments following adoption. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The estimates describe equilibrium timing; concentrated-liquidity venues remain outside this capital measure.

5 Capital, exact prices, and route use

Once both vehicle families are executable, current exact prices and lagged full-range capital jointly predict vehicle use. Full-range capital captures one part of pool state; size-specific pretrade quotes determine what routers can execute. For each observed two-leg route on the fifteenth day of every month from June 2020 through June 2026, we quote the best WETH path and the best stablecoin path through DAI, USDC, or USDT at the same input amount and pre-transaction state. The comparison covers Uniswap v2, SushiSwap v2, and Uniswap v3, which carry 93.4% of linear routes on the sampled dates. Among routes for which all required states are observed, 89.2% reproduce realised output within one basis point (Appendix Table 14, panel A). Both vehicle families must be feasible, each observed and hypothetical leg remains below 5% own-price impact, leg values agree within 20%, and input value is at least \$100. Exact price is central to automated routing, while gas, venue access, and support can still separate an available quote from an executable path (Xi and Moallemi, 2026). Appendix B gives the pricing equations, Appendix D defines the common-support rule, and Appendix Table 14 reports results for additional venues, vehicles, and direct paths.

We call a vehicle family the incumbent when it alone carries a pair on the first observed vehicle day and begin measuring retention 30 days later. For route r, pair p, and date t, we estimate

$$R_{rpt} = \beta_O A^O_{rpt} + \beta_K A^K_{pt-1} + \beta_q \log q_r + \alpha_p + \delta_t + \varepsilon_{rpt}. \tag{4}$$

Here R_{rpt} indicates retention of the incumbent family and q_r is the route's observed dollar input value. We define the two advantages as

$$A_{rpt}^{O} = \frac{1}{100}\left[10^4 \frac{O_{rpt}^{\text{inc}} - O_{rpt}^{\text{chall}}}{O_{rpt}^{\text{chall}}}\right]_{-1000}^{1000},$$

$$K_{pt-1}^{v} = \min\{K_{i,v,t-1}, K_{v,j,t-1}\},$$

$$A_{pt-1}^{K} = \frac{K_{pt-1}^{\text{inc}}/(K_{pt-1}^{\text{inc}} + K_{pt-1}^{\text{chall}}) - 1/2}{0.10}.$$

One unit of A_{rpt}^{O} represents 100 basis points. Each leg of K_{pt-1}^{v} aggregates prior-calendar deposited capital across full-range Uniswap v2 and SushiSwap v2 pools. One unit of A_{pt-1}^{K} represents a 10 pp shift from an equal capital split. Pair effects α_p absorb persistent routing conditions, and date effects δ_t absorb market-wide conditions.

Table 6, panel A, reports Equation (4). On the common sample in columns 1 and 2, a 100 basis point incumbent output advantage predicts +10.56 pp higher retention before controlling for prior capital and +10.13 pp after controlling for it. A 10 pp incumbent capital-share advantage predicts another +2.77 pp in column 2. The exact-output coefficient remains similar across columns. Current quotes predict route choice, while prior two-leg capital contains additional information about the incumbent's position.

Table 6: Capital, exact prices, and route allocation

Panel A. Route retention inside established endpoint pairs		
Outcome; estimates [pp]	(1) Incumbent retained	(2) Incumbent retained
Incumbent exact-output advantage [100 bp]	$+10.56^{***}$ (0.83)	$+10.13^{***}$ (0.77)
Incumbent prior full-range capital-share advantage [10 pp]		$+2.77^{***}$ (0.61)
Dependent mean [%]	69.7	69.7
Within R^2	0.145	0.153
Observations	17,778	17,778
Pair clusters	678	678
Pair and date fixed effects	Yes	Yes
Panel B. Route allocation when exact-price leadership changes		
Outcome; estimates [pp]	(1) Challenger leads next month	(2) Incumbent-share change
Challenger prior weak-leg capital share [10 pp]	$+3.70^{**}$ (1.52)	
Exact-price crossing [vs. the same event's earlier movement]		-28.98^{***} (4.71)
Observations	286	440
Ordered endpoint pairs	122	83
Crossing-month fixed effects	Yes	No
Crossing-event fixed effects	No	Yes

Note: Panel A uses realised two-leg routes on the fifteenth day of each month, at least 30 days after a pair first used one vehicle family exclusively. For the same pair, input amount, and pre-transaction state, we quote the best WETH path and the best path through DAI, USDC, or USDT across Uniswap v2, SushiSwap v2, and Uniswap v3. Input value is at least $100, both vehicle families are feasible, and every observed and quoted leg remains below 5% own-price impact. Both columns use routes with positive prior-day full-range capital for each family and estimate Equation (4). Weak-leg capital is the smaller deposited-capital stock across the two required legs in Uniswap v2 and SushiSwap v2. Panel B uses price-rank crossings with at least two common-support routes and $1,000 of observed input in both crossing months. Column 1 estimates the first line of Equation (5), with standard errors clustered by pair and calendar month. Column 2 estimates the second line with crossing-event effects and pair-clustered standard errors. Coefficients and standard errors are in percentage points. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. Deposited capital is an equilibrium pool stock, and exact prices cover the stated public venues and observed trade sizes.

We then follow the same pair when exact-price leadership changes. A crossing occurs when the pair-month median exact-output gap moves from at least one basis point in favour of the incumbent in month $t-1$ to at least one basis point in favour of the challenger in month t. Let $L_{e,1}$ indicate that the challenger remains the price leader one month later, and let $Q_{e,-1}$ be its prior-calendar weak-leg capital share. We estimate

$$\begin{aligned} L_{e,1} &= a + \theta Q_{e,-1} + X_e' c + \lambda_{m(e)} + u_e, \\ \Delta R_{e\tau} &= \mu_e + \rho\,\mathbf{1}\{\tau = 0\} + Z_{e\tau}' d + \nu_{e\tau}, \end{aligned} \tag{5}$$

where the second line stacks the crossing-month change from $t-1$ to t with the same event's earlier change from $t-3$ to $t-2$. Event effects μ_e compare the two changes within a crossing. The first line controls for challenger family, the price gaps in months $t-1$ and t, prior incumbent route share, log route activity, and crossing-calendar-month effects. In the second line, $Z_{e\tau}$ contains interactions of the crossing-month indicator with prior challenger capital share and an indicator for a stablecoin challenger.

Incumbent route share falls 29.0 pp more at the crossing than during the same event's earlier movement (Table 6, panel B, column 2). A 10 pp larger challenger share of prior weak-leg capital predicts 3.70 pp greater probability that the new price lead lasts one month (panel B, column 1). Route share moves with the exact-output ranking; prior two-leg capital predicts whether the challenger keeps the lead.

The first sampled contest arrives slowly for most entering pairs that reach one. Only 0.5% of material entrants enter the monthly common-support sample, after a median of 115.5 days; the entry family carries 83.1% of routes at that first contest (Appendix Table 37, panel A). Much of the persistence documented in Section 3.3 therefore accumulates before a sampled rival becomes feasible. At the first contest and in the mature sample above, current exact prices strongly predict the chosen family.

Persistence has a modest average execution cost in the common-support sample. The observed vehicle family offers at least one basis point less output on 12.9% of contestable routes, with an input-value-weighted shortfall of 7.4 bp (Appendix Table 40, panel A). Among routes that retain a mature exclusive incumbent, the corresponding incidence is 10.7% and the weighted shortfall is 3.3 bp (panel B). Incorporating predicted path-specific gas changes the overall weighted shortfall from 7.46 to 7.65 basis points (panel D). The same table shows that the shortfall is concentrated in younger and smaller pairs.

Appendix Table 37 studies the first sampled date on which both families become feasible. Appendix Tables 38 and 39 report the complete route-choice and crossing specifications, while Appendix E records the venues and pricing models covered.

6 Discussion and policy implications

We group stablecoins by their shared vehicle function while retaining the token that supplies each route. USDC, USDT, DAI, and other stablecoins target a common unit of account while differing in issuer, backing, redemption, peg risk, and local pool liquidity. Peg stress can reallocate trading across issuers (Liu et al., 2026; Anadu et al., 2025). Our family measure captures tokenised-dollar intermediation; the issuer results show how that role is divided within the family. Stable-to-stable endpoint pairs make this distinction especially clear because USDT supplies most of their increase in routed value (Appendix Table 25, panel B).

A one-for-one peg aligns prices while preserving currency identity. The Bahamian dollar, for example, is conventionally pegged at one Bahamian dollar per U.S. dollar and remains a separately issued currency with its own monetary and exchange-control arrangements (International Monetary Fund, 2025). The same distinction guides our stablecoin aggregation. A unit-of-account measure would consolidate dollar-pegged instruments. Our route measure records competition among separately issued dollar instruments and between that family and WETH.

The provider results suggest that vehicle networks reproduce themselves through specialised participation. Origins that supplied a vehicle across earlier endpoint pools are more likely to supply that vehicle when a new spoke becomes material. The stablecoin-core estimates are consistent with liquidity in DAI, USDC, and USDT core pools extending into new endpoint pools, although their uncertainty is wider. A newly issued stablecoin may gain convertibility through links to incumbent dollar tokens, and issuer-sponsored or affiliated market makers may seed those links alongside independent providers. Transaction origins reveal participation. Provider identity, affiliation, funding relationships, hedges, fee expectations, and beneficial ownership remain outside the observed records.

Demand for dollar-linked assets offers another interpretation of the upstream allocation. Demand for a stable store of value can raise expected stablecoin trading and fee income, encouraging liquidity provision on stablecoin legs (Catalini et al., 2022; Lyons and Viswanath-Natraj, 2023). Traditional models similarly connect demand for safe dollar assets and trade finance to the currency's durable international role (Gopinath and Stein, 2021; Chahrour and Valchev, 2022). Monthly growth in circulating stablecoin supply provides a broad equilibrium proxy for this demand. Its four coefficients for later core or spoke capital growth and first-link formation are imprecise after Holm adjustment (Appendix Table 29, panels A and B). Circulating supply combines issuance, redemption, transaction demand, and portfolio demand. It is too coarse to distinguish user demand from issuer support.

Aggregate stress can change market-making capacity in traditional markets (Brunnermeier and Pedersen, 2009; Nagel, 2012). In Uniswap v3, the stable-minus-WETH slopes on gross additions and withdrawals are nearly identical and individually imprecise after Holm adjustment; the net coefficient is +0.0003 (Appendix Table 35, panel A, column 1). The coefficient on an ETH-price fall is +0.0027 with Holm-adjusted $p = 0.067$, while its addition and withdrawal components are also imprecise (column 2). These coefficients compare stablecoin-facing with WETH-facing decoded external additions, withdrawals, and net flows at the pool-week level. Market-wide activity, within-pool range changes, and adjustments between external actions are separate.

Price movements create a second margin between provider actions. An ETH-price move can open a cross-rate gap between an endpoint–WETH pool and an endpoint–stablecoin pool. When that gap exceeds fees and other trading costs, triangular arbitrage through the endpoint shifts reserve ratios toward a common cross-price (Daian et al., 2020). Dollar capital across the two pools remains a separate object. Section 4.2 shows that stablecoin-facing constant-product capital rises by about 4–5% relative to WETH-facing capital over one to seven days. The unit-value component exceeds the total point estimate, while the relative $\sqrt{k}$ component is negative and imprecise after Holm adjustment (Appendix Table 36, columns 1–3). This fixed-pool accounting comparison leaves convergence speed open; measuring it requires transaction-level deviations between pool cross-rates and outside prices, and cross-market segmentation can slow convergence (Makarov and Schoar, 2020). At the monthly horizon, route-specific weak-leg capital, exact-output advantage, and realised stablecoin use have near-zero ETH-return coefficients (Appendix Table 35, panel B, row 1). We therefore find short-horizon revaluation of relative dollar capital and no detected monthly route-use response. Direct measures of stable-asset demand remain necessary to separate prospective fee demand from issuer support and provider specialisation.

Relative-price risk can also influence where capital is supplied. While two stablecoins hold their pegs, their relative price typically varies less than either token's price against WETH. This low-risk argument applies most directly to stablecoin-core pools; endpoint–stablecoin spokes still bear endpoint-token price risk. In a constant-product pool, smaller relative-price movements reduce loss versus holding, commonly called impermanent or divergence loss. Lower prior endpoint–vehicle risk is associated with deeper constant-product bridge capital at the measurement date and 30 days later (Appendix Table 34, panel A, columns 1–4). Stablecoin bridges nevertheless have higher median relative volatility than WETH bridges and carry the lower risk in fewer than one third of comparable pair-months (panel B). The vehicle comparison points to expected fees, hedging, issuer support, and network experience for the aggregate rotation. Impermanent loss is distinct from loss-versus-rebalancing, the running adverse-selection cost of stale AMM prices (Milionis et al., 2022).

Vehicle use connects otherwise unrelated pairs to the liquidity and issuer conditions of the asset in the middle. Concentration can lower trading costs through deeper two-leg markets, while fragmentation can raise price impact (Chen and Duffie, 2021). The same concentration transmits disruption across many endpoint pairs. This is a network form of commonality in liquidity (Chordia et al., 2000). Payment-system design faces the same extensive margin. Nexus assigns foreign-exchange conversion to a provider, and Rialto uses a third currency when direct exchange is unavailable (Bank for International Settlements Innovation Hub, 2024, 2025). Access policy can affect both the price quoted inside an established relationship and which two-sided markets form early enough to become the bridge for new ones. Welfare analysis would additionally require user surplus, provider returns, and the incidence of vehicle-specific risk.

7 Conclusion

Vehicle dominance can rotate even when established trading relationships show almost no net substitution. Stablecoin use rises sharply from 2024 H1 to 2026 H1 because activity shifts toward pairs that already use stablecoins and because new pairs first appear using stablecoin routes. Movements in both directions remain substantial within continuing pairs, yet they nearly cancel in the aggregate. The vehicle present when a pair enters then predicts its later vehicle mix.

Gross stable-facing liquidity additions rotate with the trading network. Stable-facing additions rise on both an action and dollar basis, period-specific transaction origins account for much of the change, and experienced vehicle suppliers are more likely to appear when new pools become material. Before a pair first uses a stablecoin, measured capital on both required legs is associated with a +1.84 pp higher weekly probability of first use; conditional on support, a tenfold increase in stablecoin relative to WETH weak-leg capital adds +1.84

pp. In established pairs, relative capital predicts route allocation and the durability of exact-price leadership. Together, these results associate later vehicle use with the network of funded endpoint–vehicle links.

Exact-output rankings predict realised use, while prior capital on both legs predicts whether the price lead persists. Providers choose funded links and liquidity units; prices revalue standing positions between those actions. Together, the estimates are consistent with vehicle dominance forming through funded networks and persisting when those networks deliver competitive quotes. Contestable routes retain the lower-output vehicle only occasionally, and their input-value-weighted shortfall is modest on average. A deeper common network can lower exchange costs; the same network can also spread vehicle-specific risks across otherwise unrelated trading relationships.

A From contract events to economic routes

Route construction begins with the contract and event format of each exchange. We identify pools from the protocol's factory or registry, map contract addresses to tokens, and convert raw integer amounts using the token precision that applied at the time of the trade. Contract addresses define token identity because unrelated tokens can share a symbol and the same economic asset can appear in native and wrapped form. ETH and WETH are consolidated only after this address-level reconstruction; every other token remains distinct unless a stated economic classification combines it. Within each transaction, pool-swap legs are ordered by block, transaction, and log position. The resulting token flows reveal the exchange that occurred. A single connected chain gives one route and therefore one exchange between its pair endpoints; parallel chains identify an order split, and disconnected event groups remain separate. Because one transaction can contain unrelated calls as well as trades across several exchanges, this procedure also prevents two simultaneous routes from being mistaken for one longer route. Pool identities enter reconstruction; the unified route panel retains the ordered venue and token sequence together with the pair. Sender and router labels are retained separately.

Disconnected event groups become more common late in the sample. They comprise 0.8% of reconstructed components in 2024 and 6.2% in 2026. In 2026 the rate is 9.4% among components touching Uniswap v4 and 4.7% among other components. Table 7 uses the internally connected token flow of each group as a separate route. This alternative preserves the principal economic result: the 2026 stablecoin share is 43.5% by count and 77.3% by value, both slightly above the connected-transaction estimates.

Table 7: Disconnected transaction components and vehicle dominance

Sample boundary or estimate	2024	2025	2026
Panel A. Share of reconstructed components excluded			
All components	0.8%	3.7%	6.2%
Components touching Uniswap v4		11.6%	9.4%
Other components		2.7%	4.7%
Panel B. Stablecoin share among native and stable two-leg vehicles			
	2024	2026	Change [pp] (s.e.)
Connected routes, count	16.9%	42.1%	+25.2 (1.04)
Connected routes, value (20% agreement)	32.7%	76.5%	+43.8 (2.01)
Each valid component, count	16.9%	43.5%	+26.6 (1.05)
Each valid component, value (20% agreement)	32.9%	77.3%	+44.4 (1.93)

Note: Panel A reports the share of reconstructed components belonging to transactions with more than one disconnected token-flow group. Uniswap v4 begins during 2025, so its 2024 cell is empty. Panel B compares the principal connected-transaction rule with an alternative that treats each internally well-formed component as a separate route. Stablecoin shares use native and stable two-leg vehicles as the denominator and equal-weight matched January through June dates in 2024 and 2026. Value estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth. The alternative assigns no unobserved handoff between components.

Table 8 validates route construction against Ethereum logs for 206 venue-days on 74 dates and checks Fluid route labels against transaction receipts and historical pool constants.

We compare every provider swap with the independently queried logs. The comparison can add a chain event absent from the provider data, remove a provider event absent from the chain, or replace its payload or log position. We then rerun route construction on the 50 sampled dates with at least one swap correction. Available indexed records omit 35 historical auxiliary venue-days. The comparison covers the sampled dates and the connected pool sequence observed in each transaction.

A complete token sequence is sufficient for the route-count sample. Value weighting additionally requires coherent amounts on every leg and a common dollar valuation. Routes without a coherent dollar valuation remain in counts and drop from value weighting. Appendix F gives the separate exclusion for self-returning routes.

A hypothetical AAVE–WETH–USDC–UNI route makes the units concrete. Suppose one AAVE is worth $100, one WETH $2,500, one USDC $1, and one UNI $10. If one AAVE

Table 8: Ethereum-log checks and route-construction consequences

Panel A. Swap events against full-day Ethereum logs

Venue	Days	Provider swaps	Provider only	Chain only	Precision [%]	Recall [%]
Uniswap v2	74	8,837,588	0	48	100.0000	99.9995
SushiSwap v2	70	599,721	0	0	100.0000	100.0000
Uniswap v3	62	4,444,960	7	2,957	99.9998	99.9335

Panel B. Assignments in transactions touched by a swap correction

Assignment	Linked transactions	Changed	Unchanged [%]
Endpoint pair	4,881	363	92.563
Intermediary tokens	4,881	363	92.563
Intermediary asset types	4,881	363	92.563
Leg count	4,881	376	92.297
Two-leg inclusion	4,881	206	95.780

Panel C. Stablecoin shares on corrected route dates

Measure	Provider rows [%]	Corrected rows [%]	Change [pp]
Route count	24.446	24.438	-0.008
Routed value	47.836	47.820	-0.016

Panel D. January–June sampled decomposition, 2024 to 2026 [pp]

	Route count			Routed value		
Component	Provider	Corrected	Change	Provider	Corrected	Change
Stablecoin-share change	+25.920	+25.921	+0.001	+40.566	+40.566	-0.000
Within continuing pairs	+0.078	+0.078	+0.000	+1.484	+1.484	-0.000
Reallocation across continuing pairs	+6.888	+6.888	+0.000	+14.635	+14.635	-0.000
Weight on continuing pairs	+0.895	+0.895	+0.000	-2.717	-2.717	-0.000
Pairs present in one window	+18.059	+18.060	+0.001	+27.164	+27.164	+0.000

Panel E. Fluid route labels against receipts and pool constants

Scope	Components	Legs	Receipts [%]	Pool identity [%]	Precision [%]	Wilson lower [%]
Overall	180	245	100.0	100.0	100.0	98.5
Cross-venue routes	120	123	100.0	100.0	100.0	97.0
Fluid-only routes	60	122	100.0	100.0	100.0	96.9

Note: Panel A uses full-day venue ledgers for 206 venue-days across 74 sampled dates. Provider-only events are provider rows absent from the corresponding Ethereum logs; chain-only events are Ethereum-log events absent from the provider rows. Panels B and C rerun route construction on the 50 dates with at least one swap correction. Panel D repeats the pair decomposition on the six common January–June sampled dates in each endpoint year. Panel E samples coherent Fluid-containing routes within each half-year from 2024 H2 through 2026 H1, with separate high-value and rank-spread draws for cross-venue and Fluid-only routes. We compare each labelled Fluid leg with the exact receipt event and the pool's historical token constants; native ETH and WETH are one economic asset. Precision uses legs with both records available, and the Wilson column reports its 95% lower bound. Available indexed records omit 35 historical auxiliary venue-days. Auxiliary log-order corrections would collide with another provider-ordered leg in 2 transactions; together they contain 47 legs and 2 reconstructed routes. The checks cover the sampled observations; transaction-trace recall beyond them remains unmeasured.

returns 0.04 WETH, the second leg returns 100 USDC, and the third returns 10 UNI, then $q = \$100$, $Q_r = 10$ UNI, and $O_r = \$100$. Each leg carries \$100 on both sides. The route value ν_r is also \$100 because it averages the \$100 source and destination values. In the data, the same calculation uses transaction-time prices and allows limited measurement error under the value-agreement rule.

Historical precision is unverified for 13 tokens traded in 22 constant-product pools. Reserve-reconstruction analyses omit those pools. An upper-bound check removes every route involving any of the 13 tokens: the resulting bound is 5,950 routes and \$10.91 million of route value. Only 1,036 of those routes use one of the affected tokens as an intermediary, representing \$1.284 million, or 0.00043% of value on routes whose legs can all be valued. All 13 tokens lie outside the native and stablecoin groups used in the main comparison. The exclusion affects exact reserve reconstruction; the route-composition result is unchanged.

Table 9: Protocol architecture and native-asset pairing

Measure	Estimate
Panel A. Uniswap v1: ETH intermediation required	
Reconstructed token-to-token routes [count]	217,003
Panel B. Uniswap v2: arbitrary token pools allowed	
Single-leg trades executed in WETH pools, 2026 [%]	95.5
Token combinations first traded with WETH, 2026 [%]	97.9

Note: Panel A reports connected Uniswap v1 token-to-token routes reconstructed from the exchange registry and matched ETH legs. The protocol allowed only ETH–token pools, so each such route crossed ETH. Panel B reports Uniswap v2 activity during January through June 2026. A single-leg trade uses a WETH pool when either pool token is WETH. A token combination first traded with WETH when either token in the newly active combination is WETH. V2 allowed arbitrary ERC-20 token pools throughout this period.

B Pricing models and validation

The counterfactual comparison requires an exact-input quote from each pool's state and the proposed input amount. Because pool families use different invariants and arithmetic, we validate their pricing models separately. For selected executed swaps, we reconstruct the state immediately beforehand and compare predicted with realised output. Curve's amplification coefficient and Balancer's weight ratio are unavailable in the historical subgraph records, so we estimate them on one set of trades and evaluate the resulting quotes on held-out trades. A route enters the cost comparison only when the relevant pool states and pricing models are available before the transaction.

For a route r, let $e(r)$ record its block, transaction position, and first pool-event position; $e(r)^-$ is the state immediately before that transaction begins. The route has an ordered pool sequence $(p_1, \ldots, p_{m(r)})$ and token sequence $(k_0, \ldots, k_{m(r)})$, where $k_0 = i$ is the input and $k_{m(r)} = o$ is the output. The dollar input notional q corresponds to $\Delta_i(q, e) = q/P_{i,e}$ units of the input token. Let χ_{p,e^-} denote pool p's pretrade state and let $Q_p(i \to j, \Delta_i; \chi_{p,e^-})$ denote its exact-input output in token j. Composing the pool quotes gives the route's token output and dollar value,

$$
\begin{aligned}
\Delta_{k_h} &= Q_{p_h}\left(k_{h-1} \to k_h, \Delta_{k_{h-1}}; \chi_{p_h, e(r)^-}\right), & h &= 1, \ldots, m(r), \\
Q_r(q, e(r)) &= \Delta_o, & O_r(q, e(r)) &= P_{o,e(r)} Q_r(q, e(r)).
\end{aligned}
\tag{6}
$$

Pool invariants take token quantities as arguments, so the dollar notional q and token input Δ_i remain distinct. Equation (6) describes an executed route and, after reconstructing the same pretrade state, prices a feasible alternative path. The subsections below specify Q_p for each pool family.

B.1 Constant-product pools

Uniswap v2 and SushiSwap v2 price the product of reserves (Lehar and Parlour, 2025). Let R_{i,p,e^-} and R_{o,p,e^-} denote pool p's input- and output-token reserves immediately before transaction e, Δ_i the token input, f_p the fee rate, and $\varphi_p = 1 - f_p$ the fraction of input retained for pricing. An exact-input swap moves the pool along

$$
(R_{i,p,e^-} + \varphi_p \Delta_i)(R_{o,p,e^-} - \Delta_o) = R_{i,p,e^-} R_{o,p,e^-},
$$

which inverts in closed form to

$$\Delta_o = \frac{\varphi_p \Delta_i R_{o,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i}, \qquad \varphi_p = \tfrac{997}{1000},$$

the 30 basis point fee both venues charge, applied on chain as the integer ratio above.[3]

A multi-leg route composes these one hop at a time, threading the output token of one pool into the next. The cost comparison evaluates the dollar outputs of the best direct and intermediated routes at the same state and dollar input notional,

$$10^4 \Delta C^D_{i,o,k,q,e} = 10^4 \frac{O^D_{i,o,q,e} - O^I_{i,o,k,q,e}}{O^D_{i,o,q,e}}.$$

Both outputs must be computed from one reconstructed pre-transaction state. A realised trade supplies the output of the chosen route alone; the direct and intermediary alternatives require reconstructed states for every pool in the comparison. Quoting each executed constant-product swap against the state immediately before it gives a median absolute error of 0.0000% across 8,024 swaps; 96.7% of quotes are within 1% and 95.2% are within 0.01% of realised output.

B.2 Concentrated liquidity

Uniswap v3 distributes liquidity across price ranges, so executable depth depends on the active range and on the initialized ticks reached by a trade (Adams et al., 2021). Within one active range, let $\widetilde{R}_{i,p,e^-}$ and $\widetilde{R}_{o,p,e^-}$ denote the virtual reserves of the input and output assets. Liquidity L_{p,e^-} and the output-per-input marginal price ρ_{p,e^-} satisfy

$$\widetilde{R}_{i,p,e^-} = \frac{L_{p,e^-}}{\sqrt{\rho_{p,e^-}}}, \qquad \widetilde{R}_{o,p,e^-} = L_{p,e^-}\sqrt{\rho_{p,e^-}}.$$

[3] The SushiSwap v2 pool contract implements the same adjustment; see `UniswapV2Pair.sol`.

If f_p is the pool fee and $\varphi_p = 1 - f_p$, an exact input Δ_i returns

$$\Delta_o = \frac{\varphi_p \Delta_i \widetilde{R}_{o,p,e^-}}{\widetilde{R}_{i,p,e^-} + \varphi_p \Delta_i}$$

until the trade reaches the next initialized tick. Range endpoints lie on a geometric grid, and crossing tick j updates active liquidity by its signed change,

$$\sqrt{\rho(j)} = 1.0001^{j/2}, \qquad L_{p,e^-} \mapsto L_{p,e^-} \pm \Delta L_j,$$

with the sign determined by the direction of the trade. The calculation then continues from the new active range until the input is exhausted. The quoter uses the contract's integer Q96 arithmetic, rounding input amounts up and output amounts down.

The active tick map is accumulated from every mint and burn strictly before the validation day, after which same-day liquidity changes and swaps are replayed together by block and log index. Consecutive swaps in one pool supply the price state: the earlier event leaves `sqrtPriceX96` and `tick` immediately before the later swap, while the interleaved event stream supplies active liquidity. We report errors separately by input direction and tick crossing because those cases use different arithmetic branches. Table 10 applies this validation to Uniswap v4. V4 retains concentrated-liquidity pricing for static-fee pools but implements it through a singleton and flash accounting.[4] The validated static fee tiers include 0, 7, 8, and 20,000 hundredths of a basis point. Hook-bearing and dynamic-fee pools are excluded because the raw records do not identify their transaction-level cash flows and realised fees; Appendix E reports their share of v4 trading.

Every direction and tick-crossing row of Table 10 reproduces realised output within 0.001% in the Maximum column.

[4] See Adams et al., *Uniswap v4 Core*, Sections 1 and 3.

Table 10: Concentrated-liquidity quoter, by direction and tick crossing

Validation group	Swaps	Median	90th pct.	Maximum	Within 1%
token0 in, no tick crossed	2,290	0	0	9.98×10^{-4}	100.0%
token0 in, tick crossed	57	0	0	3.45×10^{-7}	100.0%
token1 in, no tick crossed	1,816	0	1.69×10^{-8}	1.48×10^{-4}	100.0%
token1 in, tick crossed	55	0	5.44×10^{-9}	8.74×10^{-9}	100.0%
Pooled	4,218	0	1.15×10^{-14}	9.98×10^{-4}	100.0%

Note: Absolute quote error in percent of realised output. The sample contains 4,218 realised Uniswap v4 swaps on three dates across 13 pools and nine token combinations, at static fee tiers 0, 7, 8, 100, 500, 10,000 and 20,000. Each quote uses liquidity events ordered strictly before the swap and the price state left by the preceding swap in the same pool.

B.3 StableSwap pools

Curve interpolates between a constant-sum curve, which is flat and charges no slippage, and the constant-product curve of Section B.1.[5] For n tokens with normalized pre-transaction balances R_{j,p,e^-}, amplification A_p, and invariant D_p,

$$A_p n^n \sum_j R_{j,p,e^-} + D_p = A_p D_p n^n + \frac{D_p^{n+1}}{n^n \prod_j R_{j,p,e^-}},$$

where $A_p \to 0$ recovers the constant product and $A_p \to \infty$ recovers the constant sum. The contract solves D_p by Newton iteration and then solves the same invariant for the post-trade output balance $R'_{o,p,e}$ after R_{i,p,e^-} is increased by Δ_i. Defining $D_\Pi = D_p \prod_j D_p/(nR_{j,p,e^-})$, the two loops are

$$D_p \leftarrow \frac{(A_p n \sum_j R_{j,p,e^-} + nD_\Pi)D_p}{(A_p n - 1)D_p + (n+1)D_\Pi}, \qquad R'_{o,p,e} \leftarrow \frac{(R'_{o,p,e})^2 + c}{2R'_{o,p,e} + b - D_p},$$

with b and c formed from the balances of the tokens the swap does not touch. Balances are rescaled to eighteen decimals before either loop runs, because the invariant treats its tokens as interchangeable at par and a six-decimal stablecoin read on an eighteen-decimal scale is

[5] See Egorov, *StableSwap—Efficient Mechanism for Stablecoin Liquidity*.

Table 11: StableSwap calibration and held-out quote error, by validation day

Day	Pools fitted	Pools excluded	Held-out trades	Median $\widehat{A}_p$	Median error	Within 1%
2021-05-28	9	14	367	12,716	0.030%	100.0%
2022-09-12	11	16	262	20,493	0.029%	100.0%
2023-12-28	20	28	471	23,428	0.039%	98.9%
2025-04-13	42	50	1,207	46,851	0.036%	99.7%

Note: The sample contains four sampled days. $\widehat{A}_p$ includes Curve's on-chain precision factor of 100. Median error is the median absolute deviation of the quote from realised output on trades the calibration never saw.

off by a factor of 10^{12}. Output is net of the pool fee,

$$\Delta_o = (R_{o,p,e^-} - R'_{o,p,e} - 1)\varphi_p,$$

carrying the contract's own off-by-one guard on the raw difference.

A_p is Curve-specific and the standardised subgraph schema drops it. We identify it from data because every other quantity in the invariant is observed and every Curve swap reports both amounts,

$$\widehat{A}_p = \arg\min_{A_p} Q_{0.90}\left\{\frac{|Q_p(i \to o, \Delta_i; \chi_{p,e^-}(A_p)) - \Delta_o|}{\Delta_o}\right\},$$

fitted on the first half of a pool-day's trades over a logarithmic grid with a ternary refinement, then evaluated on the trades not used for estimation. A pool-day is retained when the 90th percentile of fitting error is below 1%. Using the upper tail prevents a partial fit from passing: one misspecified pool produced a median fitting error below 0.1% but a 34% median error on the other trades because the invariant fit only part of its activity. Curve has also operated crypto-pools since 2021 that follow the CryptoSwap invariant. Applying StableSwap to these pools yields a best-fit A_p of 3 and a 36% median quote error, so pool-days whose observed trades are inconsistent with StableSwap remain outside this validation.

Table 11 reports the held-out fit by validation day.

Across four validation days, held-out median errors range from 0.029% to 0.039%.

B.4 Weighted pools

Balancer weighted pools use a weighted geometric-mean invariant,[6]

$$K_{w,p} = \prod_j R_{j,p,e^-}^{w_{j,p}}, \qquad \sum_j w_{j,p} = 1,$$

which gives the two-token exact-input form

$$\Delta_o = R_{o,p,e^-} \left[1 - \left(\frac{R_{i,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i} \right)^{w_{i,p}/w_{o,p}} \right].$$

Only the ratio $w_{i,p}/w_{o,p}$ enters. An equal-weight Balancer pool and a constant-product pool with the same reserves therefore return the same quote. Balancer's weighted-pool contract accepts inputs up to 30% of the input-token reserve; that contract limit governs whether a quote can execute, but it is not the empirical support rule.[7] The counterfactual design applies the common 5% own-price-impact cutoff in Equation (8) to Balancer and every other pool family. The subgraph reports the weight ratio $\theta_p = w_{i,p}/w_{o,p}$ and the swap fee at its head block while balances come from a historical snapshot, so a pool that has repriced itself since is quoted with a current parameter against a past state. Liquidity-bootstrapping pools move weights on a schedule by design and managed pools move them on demand, and a weighted pool's owner can reset the fee at will. What such a pool produces is a small and nearly constant relative bias, which appears directly in the preceding validation: reported parameters reproduce trades in 2024 to a millionth of a basis point and trades in 2022 only to 0.19%. A gap of that size is a fee. The fee enters the quote through $\varphi_p \Delta_i$, so a fee wrong by a basis point moves output by about a basis point, whereas a mistake in the invariant would leave an error that scales with trade size.

We therefore treat both quantities as potentially stale and recover replacements from

[6] See Martinelli and Mushegian, *A Non-Custodial Portfolio Manager, Liquidity Provider, and Price Sensor*.

[7] See the input-ratio bound in Balancer's `WeightedMath.sol`.

executed trades when reported parameters fail the fitting-error threshold, as with Curve's amplification coefficient in Section B.3. Reconstructing a pool's state from its own event history follows Lehar and Parlour (2025); what the weighted family adds is that two arguments of the pricing function are themselves missing from the historical record, so a reconstructed state alone does not deliver a quote. Every quantity in the weighted invariant other than θ_p and f_p is observed, and every Balancer swap reports both amounts, so with balances reconstructed either parameter is the only unknown. We order a pool-day's trades and assign them alternately to a fitting set $\mathcal{O}_p$ and an evaluation set, and every error reported in Table 12 is measured on the evaluation set. For a trial parameter pair ϑ,

$$\mathcal{E}_p(\vartheta) = Q_{0.90}\left\{ \frac{|Q_p(i \to o, \Delta_i; \chi_{p,e^-}(\vartheta)) - \Delta_o|}{\Delta_o} : e \in \mathcal{O}_p \right\}$$

is the fitting error, defined only when at least nine tenths of $\mathcal{O}_p$ returns a quote at all and undefined otherwise. Using the upper tail prevents a fit based on only one part of the pool's activity, and the coverage condition requires the parameter to price almost all fitting trades.

Each token combination's weight ratio is fitted on the trades crossing it in either direction, with a trade running from b to a scored at the reciprocal exponent, which is what the invariant implies and which doubles the trades available to pin one number,

$$\widehat{\theta}_{ab,p} = \arg\min_{\theta} \mathcal{E}_p\big(\theta \text{ on } \mathcal{O}_p^{a\to b},\ \theta^{-1} \text{ on } \mathcal{O}_p^{b\to a}\big),$$

searched over a logarithmic grid spanning Balancer's own weight bounds and refined by ternary search in the log exponent, and attempted only for token combinations carrying at least four fitting trades, since one or two are fitted exactly by construction and then read as an exact fit. The fee $\widehat{f}_p$ is recovered the same way over the fee range the vault permits. Reading is tried before fitting and the fee before the weights, so each tier adds at most one free scalar and a pool that reproduces its trades on reported parameters is never handed a fitted one,

$$\widehat{\vartheta}_p = \begin{cases} (\theta_p, f_p) & \text{if } \mathcal{E}_p(\theta_p, f_p) \le \bar{e}, \\ (\theta_p, \widehat{f}_p) & \text{else if } \mathcal{E}_p(\theta_p, \widehat{f}_p) \le \bar{e}, \\ (\widehat{\theta}_{\cdot,p}, f_p) & \text{else if } \mathcal{E}_p(\widehat{\theta}_{\cdot,p}, f_p) \le \bar{e}, \\ \text{excluded} & \text{otherwise,} \end{cases} \qquad \bar{e} = 0.1\%. \tag{7}$$

The label a pool carries in the source plays no part in this. A pool-day whose observed trades no tier reproduces is excluded whatever its `poolType` says, and the third tier further requires the fitted ratios to cover nine tenths of the fitting trades, so a pool-day is never accepted on the token combinations that happened to fit and then quoted on the rest.

The frequency with which the parameter search binds measures the sufficiency of reported pool state. Of the 256 pool-days accepted across the twelve validation days, 209 satisfy the error threshold on reported parameters with nothing identified, 43 need the swap fee alone, and 4 need per-token-combination weight ratios. Four fifths of the priced sample therefore carries no fitted parameter, and identification repairs a stale field in a minority of pool-days instead of supplying the invariant.

We set $\bar{e}$ at 0.1% instead of the 1% Curve uses. Weighted pools have held-out errors from 10^{-9} to 0.004%, while stable, composable-stable, and elliptic-curve pools have held-out errors from 0.05% to 1.08% and a 99th percentile of 12%. At a 1% threshold, fitted weight ratios would admit the second group. The 0.1% rule excludes it. Curve's failed validation arose from an over-permissive parameter range; here, an over-permissive error threshold would produce the same mistake.

Across the twelve date rows of Table 12, the Backward column returns a median error of 0.0000% and the Pools priced / excluded column identifies 256 of 473 testable pool-days as weighted pools. The other pool-days carry a median 91.8% of tested swap volume and follow different Balancer pricing rules. This validates identified weighted pools; other Balancer families use different pricing rules.

Table 12: Weighted-pool quoter and reserve-timing alternatives

Day	Held-out trades	Backward	Flat	Opening	Pools priced / excluded
2021-04-25	4	4.7×10^{-9}	3.00%	3.94%	1 / 0
2021-09-29	273	2.5×10^{-8}	1.24%	0.80%	18 / 8
2022-03-09	524	1.6×10^{-4}	1.05%	2.66%	25 / 4
2022-08-16	701	1.1×10^{-9}	0.75%	1.04%	33 / 12
2023-01-24	832	2.0×10^{-10}	3.73%	4.71%	44 / 11
2023-07-03	301	7.1×10^{-10}	1.78%	3.32%	26 / 19
2023-12-11	185	2.8×10^{-9}	1.77%	2.91%	17 / 20
2024-05-20	265	3.8×10^{-10}	4.81%	5.17%	15 / 36
2024-10-27	142	1.8×10^{-10}	1.68%	2.24%	12 / 36
2025-04-06	248	2.1×10^{-10}	5.32%	9.48%	19 / 40
2025-09-13	501	2.5×10^{-4}	1.09%	1.71%	27 / 25
2026-02-21	582	8.7×10^{-9}	0.68%	1.90%	19 / 6

Note: Median absolute quote error in percent of realised output, on trades no fit ever saw. *Backward* reconstructs per-swap balances by rolling the day's event sequence back from the closing snapshot, *Flat* keeps the snapshot balances constant across the day, and *Opening* treats the snapshot as the day's opening state and rolls forward. The sample contains twelve days from 2021-04 to 2026-02.

C Reconstructing pool balances before each trade

Every counterfactual quote uses the pool state immediately before the trade, whereas some historical records store balances only at the end of an hour or day. We recover the earlier state by ordering every balance-changing event within the period and unwinding the sequence from the recorded closing balance. We validate this procedure on selected constant-product and weighted-pool observations. The cost comparison retains only routes whose pools can be reconstructed in this way.

Let $\mathbf{R}_p^{\text{end}}$ be pool p's stored period-end reserve vector and $\boldsymbol{\Delta}_{p,e}$ the vector of net signed token amounts in event e. The state before event e is the stored value with every later event removed,

$$\mathbf{R}_{p,e}^{\text{pre}} = \mathbf{R}_p^{\text{end}} - \sum_{e' \geq e} \boldsymbol{\Delta}_{p,e'} = \mathbf{R}_{p,e+1}^{\text{pre}} - \boldsymbol{\Delta}_{p,e},$$

which is a single reverse pass over the period. The alternative reading takes the previous period's stored value as the current period's opening state and replays forward,

$$\widetilde{\mathbf{R}}^{\mathrm{pre}}_{p,e} = \mathbf{R}^{\mathrm{end}}_{p,\mathrm{prev}} + \sum_{e'<e} \mathbf{\Delta}_{p,e'},$$

which accumulates every unrecorded balance change between the two stored points. Replaying forward returns a median absolute quote error of 11.8% against realised outputs, whereas unwinding backward returns 0.0000%.

Liquidity additions and withdrawals also move constant-product pool balances; the ordered replay includes them alongside swaps. We unwind the sequence to the start of the period and compare the reconstructed opening balance with an independently observed reserve state,

$$\left\| \left(\mathbf{R}^{\mathrm{pre}}_{p,1} - \mathbf{R}^{\mathrm{end}}_{p,\mathrm{prev}} \right) \oslash \mathbf{R}^{\mathrm{end}}_{p,\mathrm{prev}} \right\|_{\infty} < \tau,$$

Agreement establishes that the event history reproduces the observed reserves for the tested period. The state-dependent sample omits a period when a required event family is incomplete or the balances do not reconcile. Coverage is reported by exchange and year because missing state can be concentrated in actively managed or newly launched pools.

The same logic applies to weighted pools. We treat `poolSnapshots.amounts` as the closing balance and unwind joins, exits, and swaps in their recorded order. Table 12 compares this reconstruction with two alternatives that either hold the closing balance fixed or treat it as the opening state.

The Balancer replay includes joins and exits as well as swaps. Omitting those events assigns intraday liquidity changes to the pricing rule and changes the pool states available for the price comparison.

D Restricting hypothetical trades to observed size–depth combinations

Executed-swap validation samples contain pools deep enough to serve the observed trade. Hypothetical routes can demand much more output relative to pool depth, where the observed errors no longer describe pricing accuracy. Let $\widetilde{Q}_\ell$ denote leg ℓ's output at the pool's marginal rate. We include a hypothetical leg only when

$$\pi_\ell(\Delta_i, e^-) = \frac{\widetilde{Q}_\ell(\Delta_i, e^-) - Q_\ell(\Delta_i, e^-)}{\widetilde{Q}_\ell(\Delta_i, e^-)} \leq \bar{\pi}, \qquad \bar{\pi} = 0.05. \tag{8}$$

The ceiling is calculated under the pricing rule of the pool family the leg would traverse; contract-specific limits add further restrictions.

For constant-product swaps, input relative to reserves provides a transparent size–depth comparison,

$$\iota_{i,p,e^-} = \frac{\Delta_i}{R_{i,p,e^-}},$$

the input amount as a fraction of the input-side reserve. Table 13 reports its distribution in the historical validation sample. The statistic helps calibrate what constitutes a large trade in a constant-product pool, but it does not replace the common own-price-impact rule for concentrated-liquidity, StableSwap, or weighted pools.

In the Pooled row of Table 13, five per cent of input reserves sits between the 90th percentile of 3.29% and the 99th of 14.86%; the same cutoff lies between those columns on seven of the eight date rows. Figure 2 shows the daily distributions. This calculation locates the cutoff in the empirical size distribution. The empirical rule remains a 5% own-price-impact cutoff computed separately under each pool family's pricing function.

Equation (8) applies the same 5% own-price-impact ceiling to every family. The pool-family validations establish pricing accuracy for realised swaps within their stated samples;

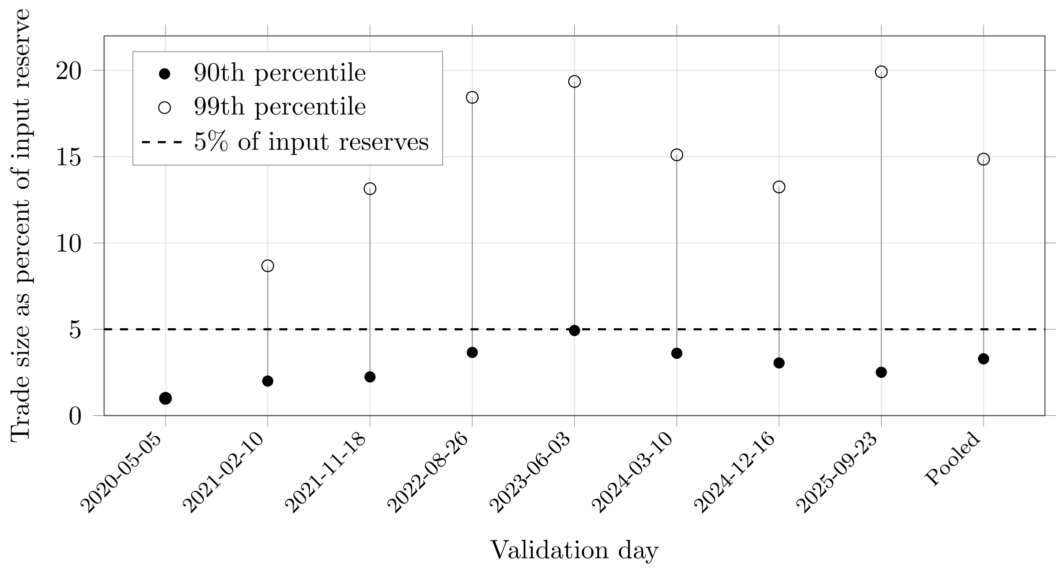
Table 13: Size relative to depth in the validation population

Day	Swaps	Median	90th pct.	99th pct.	99.9th pct.	Maximum
2020-05-05	2	1.00%	1.00%	1.00%	1.00%	1.00%
2021-02-10	141,804	0.19%	2.00%	8.68%	23.17%	99.83%
2021-11-18	107,059	0.10%	2.24%	13.15%	71.64%	100.00%
2022-08-26	79,180	0.48%	3.66%	18.44%	94.13%	100.00%
2023-06-03	159,551	0.85%	4.93%	19.36%	96.59%	100.00%
2024-03-10	215,060	0.41%	3.61%	15.11%	57.81%	100.00%
2024-12-16	132,508	0.31%	3.05%	13.25%	71.93%	100.00%
2025-09-23	97,106	0.24%	2.51%	19.92%	99.11%	100.00%
Pooled	932,270	0.34%	3.29%	14.86%	78.33%	100.00%

Note: Trade size as a percentage of the input-side reserve of the pool traversed, over the constant-product swaps the quoters were validated against. The sample contains eight dates from 2020-05-05 through 2025-09-23.

they do not measure quote error on unexecuted paths.

Figure 2: Constant-product trade size relative to reserves



Note: The statistic is a leg's input amount as a fraction of the input-side reserve in the constant-product pool it traverses. Points mark the 90th and 99th percentiles; the dashed line marks 5% of reserves. The two percentiles coincide on 2020-05-05. The sample contains 932,270 realised swaps on eight dates from 2020-05-05 through 2025-09-23.

E Coverage of the price comparison

Realised routes identify the assets that carried each trade. The price frontier holds the source i, destination o, dollar input q, and pre-transaction pool state e^- fixed. The best direct and two-leg outputs through intermediary k are

$$O^D_{i,o,q,e} = P_{o,e} \max_{p \in \mathcal{P}_{i,o,e^-}} Q_p(i \to o, \Delta_i(q,e); \chi_{p,e^-}) \,,$$

$$O^I_{i,o,k,q,e} = P_{o,e} \max_{p' \in \mathcal{P}_{k,o,e^-}} Q_{p'}\left(k \to o, \max_{p \in \mathcal{P}_{i,k,e^-}} Q_p(i \to k, \Delta_i(q,e); \chi_{p,e^-}) \,; \chi_{p',e^-}\right).$$

The vehicle-route shortfall relative to direct execution is

$$\Delta C^D_{i,o,k,q,e} = \frac{O^D_{i,o,q,e} - O^I_{i,o,k,q,e}}{O^D_{i,o,q,e}},$$

We report it in basis points as

$$10^4 \Delta C^D_{i,o,k,q,e}.$$

Xi and Moallemi (2026) compare realised and optimised allocation across parallel pools connecting the same assets. We extend that benchmark to sequential routes through an intermediary. Swaps and liquidity changes are replayed in on-chain order so every quote uses information available before the trade.

The opportunity set expands in three stages: alternative paths using the same vehicle and venue families as the observed route, paths using the same vehicle across Uniswap v2, SushiSwap v2, and Uniswap v3, and paths using any named public vehicle or the direct connection across those exchanges. Each observed two-leg route must reproduce realised output within one basis point. Leg values must agree within 20%, input value must be at least \$100, and every observed or hypothetical leg must satisfy Equation (8).

E.1 Realised gas burden

Receipt gas measures the fixed execution toll attached to adding a vehicle leg. In the executed-route sample, a stable-vehicle route in 2024 uses 94,290 more gas than the median direct route, 50.2% of the direct-route median. Evaluated at that year's median gas price and WETH price, the extra toll is \$3.24, so a trade needs about \$32.4k of notional before the extra gas is below one basis point. In 2026 the median stable-vehicle route uses 233,640 more gas than the direct-route median, while the dollar toll falls to \$0.08 and the one-basis-point notional falls to \$762. At each stable-vehicle route's own gas price, median stable-vehicle notional falls from \$832 to \$82, the median fixed toll falls from 44.3 bp to 16.2 bp, and the share of routes with a toll below 25 basis points rises from 39.5% to 55.7%. The extra leg adds gas units but becomes small relative to most observed stable-vehicle notionals. These receipt summaries describe the realised fixed toll; the path-specific model below predicts the toll for both feasible vehicle paths.

Table 14: Exact pre-transaction prices and vehicle identity

Panel A. Coverage and quote validation				
Linear routes in the exact venue set [%]				93.4
Mapped observed routes reproduced within 1 bp [%]				89.2
Panel B. Nested opportunity set				
	Realised route	Same vehicle, used venues	Same vehicle, all exact venues	Any named vehicle or direct
Routes with more than 1 bp higher output [%]	0.0	6.6	44.5	46.4
Stablecoin vehicle share [%]	19.5	19.5	19.5	18.4
Full-set minus realised stablecoin share [pp]			Route weighted	-1.16^{***} (0.19)
			Input-value weighted	-2.09^{***} (0.31)
Routes / dates				777,203 / 73

Note: The sample contains exact two-leg routes executed on Uniswap v2, SushiSwap v2, or Uniswap v3 on the fifteenth day of each month from June 2020 through June 2026. Source, destination, input amount, and pre-transaction pool state remain fixed. Each observed quote reproduces realised output within one basis point. Leg values must agree within 20%, input must be at least $100, and every observed or hypothetical leg must remain below 5% own-price impact. The public vehicle set contains WETH, USDC, USDT, DAI, and WBTC; a realised vehicle outside that set remains eligible. A route changes vehicle only when gross output improves by more than one basis point. Standard errors in parentheses are clustered by date. P-values use Holm adjustment across route- and input-value-weighted comparisons in five reported samples. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The three exact venues contain 93.4% of linear routes over the sampled dates, and 89.2% of mapped observed routes reproduce within one basis point. Coverage falls as other exchanges expand late in the sample. The comparison is gross of gas and excludes private order flow. 98 routes imply more than twice realised output under standard invariant pricing; removing them leaves every displayed share unchanged at one decimal place, and they are excluded from the median-gain calculation.

For the path-specific comparison, we fit transaction gas on 40,186 receipt-measured transactions spanning April 2019 through June 2026. Predictors are year, transaction-callee class, ordered venue sequence, route-leg count, and cross-venue status. A deterministic holdout contains 7,520 transactions. Median absolute percentage error is 17.4%, compared with 34.5% for a leg-count-only model (Table 15, panel A). The central prediction matches 52,433 of 52,477 common-support routes to receipts and retains 52,207 with positive output net of predicted gas on both paths. Panel B applies the path-specific interquartile predictions in both directions. These bounds are wide for small trades, so the central adjustment enters the main comparison and the bounds remain here.

Table 15: Gas prediction and path-specific output bounds

Panel A. Held-out prediction accuracy

Sample	Test routes	Actual gas	MAE	MAPE [%]	Legs-only MAPE [%]	IQR coverage [%]
All routes	7,520	290,864	49,762	17.4	34.5	45.0
Two-leg routes	4,860	289,920	49,651	17.4	35.1	46.0

Panel B. Path-specific interquartile gas bounds

	Lower-output route [%]			Weighted shortfall [bp]		
Input value	Central	Favorable	Unfavorable	Central	Favorable	Unfavorable
All routes	13.7	6.2	30.1	7.65	6.31	11.85
$100–999	14.7	4.2	41.7	22.01	12.01	104.67
$1,000–9,999	13.0	6.5	24.1	9.51	6.46	18.48
$10,000–99,999	13.1	10.5	16.5	9.21	8.04	10.83
$100,000 and above	9.7	9.7	10.8	3.77	3.68	3.89

Note: Panel A uses a deterministic transaction-hash holdout from the 40,186-transaction receipt panel. Actual gas, median absolute error (MAE), and prediction features refer to total successful-transaction gas units, including router transfers and bookkeeping. The legs-only benchmark uses route-leg count alone. Interquartile-range (IQR) coverage is the share of realised receipt gas between the cell-specific 25th and 75th percentiles. Panel B subtracts path-specific gas from the exact stablecoin and WETH outputs using the executed transaction's effective gas price and same-day WETH price. The central estimate uses cell medians. The favorable bound assigns the observed route its 25th-percentile gas and the rival path its 75th percentile; the unfavorable bound reverses them. The rival path retains the observed transaction-callee class. A shortfall starts above one basis point, and weighted means assign zero below that threshold.

E.2 Exchange and pool-family coverage

Omitting an exchange can change the best direct route, the best intermediated route, or both. Panel A of Table 16 describes the volume recorded in nine local source series, including Uniswap v1 and the available Fluid extracts. Volume in the Graph-indexed series is reconstructed from individual swaps using the smaller dollar value of the two legs. This avoids double counting in two series and removes implausible oracle values in thin pools: one Curve pool reports a $456m day whose swaps sum to $8, and a constant-product pool with $255 of reserves reports $1.36bn. Physical bounds on volume relative to reserves exclude between 0.10% and 0.24% of Graph-indexed pool-days per year. Uniswap v1 enters the full route panel but not the route-cost opportunity set because the price comparison requires a common pre-transaction state model. Fluid supplies volume on only 40 of the 400 sampled dates and no historical TVL field. Panel B uses DeFiLlama's daily protocol breakdown to compare the selected exchange families with total Ethereum DEX volume. Their combined share declines from 98.5% in 2020 to 77.5% in 2026 H1 and is 87.5% over the displayed

Table 16: Exchange-source composition and Ethereum market coverage

Panel A: shares within the nine retained source series									
Year	Uni V1	Uni V2	Uni V3	Uni V4	Sushi V2	Sushi V3	Curve	Balancer	Fluid
2020	2.09	75.89	0.00	0.00	10.89	0.00	11.12	0.00	0.00
2021	0.02	31.27	37.97	0.00	17.31	0.00	12.25	1.18	0.00
2022	0.01	7.01	64.07	0.00	4.35	0.00	18.70	5.87	0.00
2023	0.01	9.26	66.81	0.00	1.21	0.03	13.88	8.80	0.00
2024	0.00	11.63	69.06	0.00	0.49	0.01	12.57	5.86	0.38
2025	0.01	4.21	53.10	19.53	0.28	0.01	9.68	1.49	11.70
2026	0.01	2.27	46.02	31.94	0.11	0.16	12.61	0.15	6.73
Pooled	0.05	14.58	54.54	5.30	5.73	0.02	13.37	3.82	2.61

Panel B: selected exchange families as a share of total Ethereum DEX volume								
Period	2020	2021	2022	2023	2024	2025	2026 H1	Pooled
Share (%)	98.5	92.2	92.3	86.6	87.2	82.4	77.5	87.5

Note: Panel A reports percentage shares of volume recorded in nine local source series. Its weekly calendar contains 400 dates; Fluid is observed on 40 (6 in 2024, 25 in 2025, and 9 in 2026) and supplies no TVL field. Panel B sums the protocol families represented in the route panel and divides by total Ethereum DEX volume in DeFiLlama's daily breakdown; the numerator includes Uniswap v1–v4, SushiSwap v2 and v3, Curve, Balancer v2, and Fluid. DeFiLlama's protocol taxonomy need not match the exact transaction-level deployments in Panel A. The 2026 cell ends on June 30, and the pooled cell sums 2020 through 2026 H1. See DeFiLlama's DEX-volume series.

period. This external comparison describes the families' market reach; it does not establish transaction-level or pretrade-state coverage for every deployment.

The Uniswap v4 pricing model covers static-fee pools without hooks. Pool-identity reconstruction leaves zero swaps with incomplete or invalid immutable pool attributes across 29,005,863 rows. Pools in this group account for 84.02% of swaps and 99.05% of reported value; the corresponding 2026 shares are 79.89% and 95.57%. Other pools use hooks, dynamic fees, or both, whose transaction-level cash flows are not identified from the raw event. These shares describe the reach of the pricing model. They do not establish the availability of every pre-transaction state.

Curve carries between 9.68% and 18.70% of observed source volume in every year it exists and ranks second in four years, making its coverage economically important. The StableSwap model fits 621 pool-days and excludes 588 whose 90th-percentile fitting error exceeds 1%. Across 28 sampled dates from 2020-02-11 to 2026-04-28, these exclusions account for 34.2% of the Curve volume assessed, or \$1.83bn of \$5.36bn. Another 2.75% of Curve volume

Table 17: Curve volume outside the StableSwap comparison

Panel A: composition of the exclusion					
Leg served	Pool-days	Volume	Of excluded	Of priced	Of that leg
Native	374	$1.038bn	56.6%	15.7%	65.2%
Stable	119	$0.752bn	41.0%	79.9%	21.1%
Other volatile	95	$0.045bn	2.4%	4.4%	22.2%
Panel B: excluded share of each leg's volume, by year					
Year		Native leg		Stable leg	Other volatile
2021		85.49%		46.91%	15.23%
2022		68.82%		30.21%	50.77%
2023		59.11%		5.51%	56.60%
2024		48.24%		0.94%	14.38%
2025		47.47%		13.96%	10.52%
2026		60.92%		9.04%	20.08%

Note: A pool serves the *native* leg when it contains ETH or a staking derivative, the *stable* leg when every token is stable, including baskets and interest-bearing wrappers, and the *other volatile* leg otherwise. Stablecoin base-pool tokens are classified with the underlying stable assets. The sample contains 28 measured Curve pool-days from 11 February 2020 through 28 April 2026.

occurs in pool-days with too few trades to estimate and evaluate the amplification coefficient separately. Lowering the minimum from eight trades to four changes annual excluded shares by at most 1.3 pp. Table 17 shows where these exclusions occur.

The exclusions are concentrated in routes involving the native asset. In Panel B of Table 17, the excluded share of native-leg volume exceeds the stable-leg share by at least 33 pp in every sample year and by 54 pp in the 2023 row. Tricrypto pools pairing WBTC and WETH with a stablecoin account for $818m, or 44.6%, of the $1.83bn excluded, including seven of the twelve largest excluded pool-days. These CryptoSwap pools concentrate the missing coverage where Curve supplies depth between ETH and stablecoins.

SushiSwap v3 is omitted from the route-cost comparison because its historical records lack the pool's root price, in-range liquidity, and tick spacing. Its reserves are distributed across ticks, and the balance and weight fields exposed by the common schema do not describe that invariant. The exchange accounts for 0.016% of volume among the seven exchanges used for route costs over the full sample and 0.176% in 2026. Across fourteen sampled days, only

4.1% of its volume occurs on token combinations unavailable on another included exchange, never more than $37,609 on one day.

Balancer accounts for 3.9% of measured venue volume over the full sample and 8.8% at its 2023 peak. Table 12 validates selected weighted pools. Linear and boosted pools remain outside the quoter. Of 29 sampled composable-stable pools, 23 reduce to two external tokens after removing the pool's own share token, but pricing them also requires historical rate-provider states absent from the source. The priced Balancer sample is therefore confined to weighted pools whose historical state can be reconstructed and whose realised trades are reproduced at one of the tiers of Equation (7).

F Round-trip exclusion

A route whose first input token equals its last output token,

$$\mathbf{1}_{\{k_0=k_{m(r)}\}},$$

returns to its starting asset. We exclude these routes from the dominance measures. They include cyclic arbitrage and other self-returning activity; the endpoint rule alone leaves trader intent unidentified. Multi-trade arbitrage can be bundled within one transaction (Daian et al., 2020).

Table 18 reports the distribution across every day with at least 200 multi-leg routes.

At the median, round trips account for 12.8% of multi-leg routes by count and 21.0% by value. At the 95th percentile the shares are 23.0% and 51.1%, and the value-share maximum reaches 97.7% on 13 October 2023. The wider upper tail by value is consistent with large self-returning routes carrying disproportionate notional. Round trips are excluded before the count- and value-weighted dominance measures are formed.

Table 18: Round-trip share of multi-leg routing across all eligible days

Quantile across days	Share by count	Share by value
Minimum	0.00%	0.00%
25th percentile	9.66%	15.44%
Median	12.82%	20.96%
75th percentile	16.28%	27.99%
95th percentile	22.97%	51.07%
Maximum	33.09%	97.73%
2019 median	0.00%	0.00%
2020 median	12.85%	12.77%
2021 median	13.39%	23.95%
2022 median	14.17%	25.82%
2023 median	9.18%	17.07%
2024 median	10.38%	19.83%
2025 median	16.21%	30.47%
2026 median	21.21%	20.08%

Note: Round trips as a share of multi-leg routes. The sample contains all 2,449 eligible dates from 15 May 2019 through 30 June 2026 and 359,631,951 routes, of which 68,927,053 are multi-leg. Dates with fewer than 200 multi-leg routes are omitted.

G Additional composition and network results

For route r, let $\mathcal{M}(r)$ denote the assets between its source and destination and define $Z_{r,k} = \mathbf{1}_{\{k\in\mathcal{M}(r)\}}$. Let $a(k)$ assign each token to an economic asset group and set $Z_{r,g} = \sum_{k:a(k)=g} Z_{r,k}$ for $g \in \{\text{native}, \text{stable}\}$. On an exact two-leg route, $Z_{r,g}$ is an indicator. Let $\mathcal{R}_t^{(2),NS}$ contain the exact two-leg routes on day t whose intermediary is native or stable, and let $\mathcal{R}_t^{(2),NS,V}$ retain the routes whose source, intermediary, and destination values agree within 20%. The count- and dollar-weighted shares are

$$S_{g,t}^{N} = \frac{\sum_{r\in\mathcal{R}_t^{(2),NS}} Z_{r,g}}{|\mathcal{R}_t^{(2),NS}|}, \qquad S_{g,t}^{V} = \frac{\sum_{r\in\mathcal{R}_t^{(2),NS,V}} \nu_r Z_{r,g}}{\sum_{r\in\mathcal{R}_t^{(2),NS,V}} \nu_r}, \tag{9}$$

where ν_r is the arithmetic mean of the route's total dollar value at its observed source and destination. The value sample also requires each intermediary's incoming and outgoing dollar values to have a ratio between 0.8 and 1.2. The native and stable groups exhaust the

Table 19: Stablecoin share among native and stable intermediaries

Stablecoin share among native and stable intermediaries	2024	2026	Change [pp] (s.e.)
Route count	16.9%	42.1%	+25.2 (1.04)
Dollar-weighted routes (20% agreement)	32.7%	76.5%	+43.8 (2.01)

Note: The denominator contains exact two-leg routes whose sole intermediary is native or stable. Daily shares receive equal weight over matched January–June dates in 2024 and 2026. Dollar-weighted estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth.

denominator in Equation (9).

Let W_y be the share of native-plus-stable activity in continuing pairs, $\omega_{p,y}$ pair p's weight within that activity, and $s_{p,y}$ its stablecoin share. Continuing pairs have stablecoin share $S_{C,y} = \sum_p \omega_{p,y} s_{p,y}$, while period-specific pairs have share $S_{E,y}$. Total stablecoin share is $S_y = W_y S_{C,y} + (1 - W_y) S_{E,y}$. Writing $\Delta x = x_{2026} - x_{2024}$ and $\bar{x} = (x_{2024} + x_{2026})/2$ gives

$$\Delta S = \underbrace{\bar{W} \sum_p \bar{\omega}_p \Delta s_p}_{\text{switching within continuing pairs}} + \underbrace{\bar{W} \sum_p \bar{s}_p \Delta \omega_p}_{\text{activity shifting across pairs}} + \underbrace{(\bar{S}_C - \bar{S}_E)\Delta W}_{\text{continuing-pair weight}} + \underbrace{(1 - \bar{W})\Delta S_E}_{\text{period-specific pairs}}. \tag{10}$$

The midpoint averages hold the companion quantity fixed while each component changes; the four terms add exactly to the aggregate change.

For route-count or routed-value stablecoin share $S^{(m)}_{pds,y}$, the matched comparison is

$$S^{(m)}_{pds,y} = \alpha^{(m)}_{pds} + \beta^{(m)} \mathbf{1}_{\{y=2026\}} + \varepsilon^{(m)}_{pds,y}, \qquad m \in \{N, V\}, \tag{11}$$

where d is the month-day and s indicates whether the two legs use one exchange or several. Observations are weighted by native-plus-stable route count or supported routed value, and standard errors cluster by pair and date.

Table 20: USDT intermediary use relative to endpoint demand

	2024	2026
Count excess-use ratio (2024 full year; 2026 January–June)	1.06	1.22
Value-weighted excess-use ratio (2024 full year; 2026 January–June)	0.59	1.40
Paired January–June intermediary minus route-endpoint share [pp]	−7.13	+7.95

Note: The excess-use ratio divides USDT's share of intermediary activity by its share of route-endpoint activity on the same currency perimeter. Values above one indicate that USDT appears proportionally more often as an intermediary than as a source or destination. Count ratios use all topology-valid routes; value ratios require source, intermediary, and destination values to agree within 20%. The final row gives equal-day means over matched January–June dates.

Table 21: Pair-composition decomposition with materiality floors

	Stablecoin share [%]		Change [pp]				
Pair-period floor	2024 H1	2026 H1	Total	Within	Reweight	Support	Period-only
$\geq$ 5 routes	20.49	46.09	+25.60	−0.18	+7.38	−0.18	+18.59
$\geq$ 10 routes	21.77	47.11	+25.34	−0.24	+7.20	−0.09	+18.47
$\geq$ $5,000 value	35.75	78.63	+42.88	+0.08	+20.79	−0.38	+22.39
$\geq$ $50,000 value	37.73	80.05	+42.32	+0.61	+15.12	+0.83	+25.76

Note: Each floor is applied separately to an ordered endpoint pair in 2024 H1 and 2026 H1, using month-days present in both periods. Continuing pairs clear the floor in both periods; period-only pairs clear it once. Route floors use native- and stablecoin-mediated route counts. Value floors use routed dollars whose source, intermediary, and destination valuations agree within 20%. Within is the net stablecoin-share change inside continuing pairs; Reweight is activity shifting among them; Support is their changing aggregate weight; Period-only is the net contribution of pairs that clear the floor once. The four components sum to Total. Shares are percent; changes are percentage points.

Table 22: Pair decomposition after venue exclusions

Venue set	Stablecoin-share change [pp]					Retained activity [%]	
	Total	Within	Reweight	Support	Period-only	2024 H1	2026 H1
Panel A. Route count							
All venues	+25.48	−0.13	+8.35	−0.54	+17.79	100.0	100.0
Excluding Curve	+24.77	+0.11	+7.33	−0.56	+17.89	99.0	94.2
Excluding Uniswap v1	+25.49	−0.13	+8.35	−0.54	+17.80	100.0	99.9
Excluding Uniswap v2	+32.49	+0.03	+12.23	−0.01	+20.24	35.8	63.1
Excluding Uniswap v3	+33.43	+1.03	+7.56	−0.84	+25.68	51.2	35.1
Excluding Uniswap v4	+6.54	−1.06	+1.91	−0.15	+5.84	100.0	66.9
Excluding Balancer	+25.55	−0.11	+8.32	−0.54	+17.88	99.5	99.0
Excluding SushiSwap v2	+25.94	−0.25	+8.56	−0.57	+18.19	96.3	97.9
Excluding SushiSwap v3	+25.55	−0.10	+8.36	−0.54	+17.83	100.0	99.3
Excluding Fluid	+24.38	−0.22	+7.54	−0.53	+17.59	100.0	97.9
Uniswap v2/v3 and SushiSwap v2	+3.37	−0.82	−0.89	−0.32	+5.40	98.5	59.7
Panel B. Routed value							
All venues	+42.81	−0.03	+26.20	−2.52	+19.16	100.0	100.0
Excluding Curve	+44.44	+1.78	+29.60	−1.56	+14.63	89.2	74.9
Excluding Uniswap v1	+42.81	−0.03	+26.20	−2.52	+19.16	100.0	100.0
Excluding Uniswap v2	+36.80	+0.71	+22.68	−2.08	+15.48	66.9	92.5
Excluding Uniswap v3	+79.42	+2.01	+33.65	−1.28	+45.04	20.5	41.2
Excluding Uniswap v4	+31.34	−1.44	+20.36	−1.59	+14.01	100.0	58.9
Excluding Balancer	+42.71	−0.11	+26.22	−2.59	+19.18	98.1	99.8
Excluding SushiSwap v2	+42.62	−0.04	+26.04	−2.56	+19.19	98.9	99.8
Excluding SushiSwap v3	+42.91	+0.04	+26.24	−2.51	+19.14	100.0	99.8
Excluding Fluid	+37.44	−0.67	+23.91	−2.59	+16.78	100.0	75.6
Uniswap v2/v3 and SushiSwap v2	+21.74	−0.69	+19.83	−0.82	+3.42	87.6	33.3

Note: Each row repeats the exact midpoint decomposition over matched January–June dates after retaining only routes whose legs use the stated venue set. All venues includes the nine observed venue families. The final row retains Uniswap v2, Uniswap v3, and SushiSwap v2; it is a venue-family restriction, while Appendix Table 8 reports the separate transaction-level checks. Retained activity is the route-count or supported-value mass relative to the unrestricted sample in the same period. Within is net stablecoin-share change inside continuing pairs; Reweight is activity shifting among them; Support is their changing aggregate weight; Period-only is the contribution of pairs present in one period. The four components sum to Total.

Appendix Table 25 separates endpoint composition from intermediary identity inside stable-to-stable routes.

The full route panel supports two complementary measures. Intermediary-position share allocates all observed intermediary work across currencies. Route-participation share records the fraction of routes containing a currency and can sum above one across currencies because a longer route may contain several intermediaries. Panel A of Table 26 reports both measures for the native and stablecoin families. Panel B compares realised use with each named currency's position in the trading graph.

Table 26, panel A, shows the same stablecoin rotation under both all-route normalisa-

Table 23: Alternative accounting for the route-count rotation

Component	Estimate [pp]
Pairs entering or leaving the sample	+10.0
Pairs gaining or losing a vehicle route	−0.4
Market activity shifting across continuing pairs	+7.7
Change in how often continuing pairs use a vehicle	+6.9
Stablecoin share within continuing vehicle-using pairs	+1.3
Total route-count change	+25.5

Note: The table factors the pooled route-count change into pair support, vehicle-route support, activity reallocation, vehicle incidence, and stablecoin share within continuing vehicle-using pairs. It is an alternative accounting of the same total reported in Table 2, panel A; its components use a different mass and are not added to the main decomposition.

Table 24: Time-window and endpoint restrictions on the stablecoin rotation

Panel A. Adjacent January–June comparisons, route count

H1 comparison	Total	Within pair	Across-pair shift	Support shift	Pair entry/exit
2019–2020	+7.3	+0.0	+0.0	+3.0	+4.2
2020–2021	+11.6	+0.5	+11.3	−6.4	+6.2
2021–2022	+10.3	+0.5	+1.8	−1.5	+9.5
2022–2023	−6.4	−0.4	−1.7	−4.5	+0.1
2023–2024	−5.0	−0.4	+1.6	−6.8	+0.5
2024–2025	+8.0	−0.1	+2.9	−2.1	+7.4
2025–2026	+17.4	+1.0	+5.0	−1.1	+12.4

Panel B. Neither endpoint is WETH or a stablecoin

	2024 H1	2026 H1	Total	Within pair	Across-pair shift	Support shift	Pair entry/exit
Route count	1.1	9.0	+8.0	+0.2	+0.6	−0.0	+7.2
Supported value	0.9	39.1	+38.1	−0.3	+1.5	+3.1	+33.8

Note: Both panels apply the exact decomposition in Equation (10); endpoint-year shares are percentages, and all changes and components are in pp. Panel A uses equal-weight matched January–June dates in each adjacent pair of years. Panel B compares matched January–June dates in 2024 and 2026 after removing every pair with WETH or a stablecoin at either endpoint. Supported value requires source, intermediary, and destination values to agree within 20%. The across-pair component changes weights among continuing pairs, the support shift changes the aggregate weight on continuing pairs, and pair entry/exit is net of pairs present in only one endpoint year. Components use unrounded values and sum to the total change.

Table 25: Endpoint direction and stablecoin intermediary identity

	Route count [pp]	Routed value [pp]
Panel A. Contribution by endpoint direction		
All other pairs	+15.2	+20.6
One native, one stable endpoint	+9.2	+10.1
Two stable endpoints	+1.0	+13.2
Total stablecoin-share change	+25.4	+43.9
Panel B. Stablecoin identity within two-stable-endpoint pairs		
USDT	+0.6	+13.7
USDC, DAI, and other stablecoins	+0.4	-0.5
All stablecoin intermediaries	+1.0	+13.2
Top-decile-trimmed daily mean	–	+12.3
Pooled route-value weighting	–	+12.7

Note: Each entry is a contribution to the equal-day stablecoin-share change between matched January–June dates in 2024 and 2026. Panel A partitions the full native-or-stable intermediary sample by pair endpoints. Panel B partitions the stablecoin contribution within pairs whose two endpoints are stablecoins; its first two rows sum to the third. Routed value requires source, intermediary, and destination values to agree within 20%. The trimmed row removes the largest decile of daily contributions in each year. The pooled row weights routes by supported value across the matched dates. Components use unrounded values.

tions. Its panel B uses unweighted betweenness and ranks WETH first in every year, even as USDC and USDT gain realised intermediary use. The two-sided eigenvector measure gives greater weight to links with already central tokens and combines incoming and outgoing centrality. Table 27, panel A, shows that its count-weighted version also ranks WETH first in 2024 H1 and 2026 H1. The dollar-weighted ordering moves from WETH, USDC, and USDT in 2024 H1 to USDC, USDT, and WETH in 2026 H1.

Omitting one sampled date preserves the full-graph dollar-weighted ordering in both periods. Omitting one venue gives wider rank ranges: WETH spans ranks three to four in 2026 H1, while USDC and USDT each span ranks one to two (Table 27, panel B). After removing core stablecoin legs, WETH ranks first, USDC second, and USDT third in 2026 H1 (panel C). The dollar-weighted stablecoin lead therefore comes partly from the stablecoin core; WETH remains the leading connector outside those core legs. The issuer split in Appendix Table 25, panel B, keeps this scope dependence beside the stable-to-stable endpoint

Table 26: All-route participation and trading-network position

Panel A. All complete route lengths

Year	Currency family	Intermediary-position share [%]	Route-participation share [%]	Routes using family
2020	Native	71.7	78.2	2,915,214
	Stablecoin	25.4	26.0	970,851
2024	Native	75.3	79.5	6,894,487
	Stablecoin	17.2	17.6	1,523,281
2026 H1	Native	42.1	50.2	2,024,310
	Stablecoin	41.8	46.0	1,855,019

Panel B. Named currencies in the trading graph

Currency	Year	Betweenness (rank)	Intermediary-position share [%]	Route-participation share [%]
WETH	2020	0.965 (1)	70.7	77.2
	2024	0.988 (1)	76.2	80.5
	2026 H1	0.925 (1)	42.4	50.7
USDC	2020	0.017 (4)	10.7	11.7
	2024	0.045 (2)	8.1	8.6
	2026 H1	0.107 (3)	21.1	25.1
USDT	2020	0.096 (2)	10.7	11.7
	2024	0.027 (3)	6.8	7.1
	2026 H1	0.113 (2)	16.2	19.3

Note: Panel A uses all coherent complete routes that do not return to their starting asset. Intermediary-position shares use all intermediary positions as the denominator. Route-participation shares use complete routes as the denominator and may sum above 100% across currency families. The 2020 and 2024 rows use full calendar years; 2026 H1 covers January through June. Panel B builds one undirected trading graph for each year from legs observed on the fifteenth day of each month. An edge records at least one leg in either direction during the sampled year. Betweenness is unweighted and approximated from 256 source nodes with a fixed seed. Realised-use shares in Panel B use the same sampled dates and retain all route lengths. Ranks are shown in parentheses.

Table 27: Network-centrality rankings and coverage sensitivity

Panel A. Two-sided eigenvector centrality in the full graph

Edge weight	Currency	2024 H1		2026 H1	
		Score (%)	Rank	Score (%)	Rank
Count	WETH	8.3	1	17.4	1
Count	USDC	5.2	3	14.0	2
Count	USDT	6.0	2	12.6	3
USD value	WETH	24.0	1	15.5	3
USD value	USDC	22.1	2	31.0	1
USD value	USDT	15.7	3	30.3	2

Panel B. USD-value ranks when one sampled unit is omitted

Omitted unit	Period	WETH	USDC	USDT	Minimum Kendall τ
Sampled date	2024 H1	1	2	3	0.929
Sampled date	2026 H1	3	1	2	0.938
Venue	2024 H1	1–4	1–2	2–3	0.762
Venue	2026 H1	3–4	1–2	1–2	0.857

Panel C. 2026 H1 USD-value graph without stablecoin-to-stablecoin legs

Currency	Score (%)	Rank
WETH	25.2	1
USDC	20.4	2
USDT	13.3	3

Note: The graph contains unambiguous direct trades and every leg in a coherent multi-leg route. Count gives each observed leg equal weight; dollar value uses repriced leg value. Two-sided eigenvector centrality is the normalised geometric mean of incoming and outgoing eigenvector centrality on the largest strongly connected component. Panel B reports named-currency rank ranges and the minimum Kendall rank correlation with the full graph when one sampled date or one venue is omitted. Panel C removes stablecoin-to-stablecoin legs from the 2026 H1 dollar-value graph. Centrality depends on the edge weight and on whether the stablecoin core is included.

composition that generates it.

H Additional evidence on liquidity formation and persistence

The estimates below describe equilibrium comovement between pool formation and route use. They cover entry-activity restrictions, persistent bridge support, capital changes before first use, relative-price risk, and ETH-price movements. Provider motives require separate evidence.

H.1 Pool maturity and stable-facing liquidity additions

The increase in stable-facing liquidity additions is concentrated in mature pools. Pools older than 90 days supply 79.5% of the 68,003-action increase between 2024 H1 and 2026 H1, while pool days 0–7 supply 6.0% (Table 28, panel A). Excluding the first seven pool days, the stable-facing action share rises from 9.0% to 47.4%, or +38.40 pp (panel B). Among transaction origins present only in the 2026 H1 comparison period, one-day, one-pool participation supplies 3.58% of stable-facing actions and 0.49% of vehicle-side dollar additions (panel C). Panel D shows continued activity among pools first observed in 2026 H1. The rotation in addition activity therefore spans repeated participation in enduring pools and newly observed pools.

H.2 Stablecoin circulation and later pool formation

Table 29 relates monthly growth in each stablecoin's asset-wide circulating amount to next-month capital growth and first material links. The four core and spoke coefficients are imprecise after adjustment for their shared family. Circulating supply is an equilibrium quantity and provides a broad demand proxy; issuer support requires separate evidence.

Table 28: Pool maturity and stable-facing liquidity additions

Panel A. Stable-facing addition growth by pool age

Pool age [days]	Addition actions, 2024 H1	Addition actions, 2026 H1	Share of increase [%]
0–7	2,827	6,891	6.0
8–30	1,922	4,224	3.4
31–90	4,122	11,700	11.1
More than 90	14,371	68,430	79.5
Total	23,242	91,245	100.0

Panel B. Stable-facing share across pool-age restrictions

Pool-age restriction	Stable-facing share [%]		Change [pp]
	2024 H1	2026 H1	2024 H1–2026 H1
All pool ages	8.5	44.2	+35.71
Older than 7 days	9.0	47.4	+38.40

Panel C. Transaction-origin participation in stable-facing additions, 2026 H1

Transaction-origin history	Share of actions [%]	Share of vehicle-side USD additions [%]
Present in both comparison periods	19.54	66.02
Present only in 2026 H1; repeated days or multiple pools	76.88	33.49
Present only in 2026 H1; one day and one pool	3.58	0.49

Panel D. Later activity among stable-facing pools first observed in 2026 H1

Horizon	30 days	90 days
Pools in horizon-specific cohort	437	255
Action-weighted active-pool share [%]	83.3	69.0
Net vehicle-side USD flow / initial additions [%]	-1.22	-0.40

Note: Pool age counts days from the first retained Uniswap v3 pool-day. The sample contains spoke pools with exactly one WETH, DAI, USDC, or USDT side and an endpoint outside that set; stable-facing combines DAI, USDC, and USDT. Addition actions are positive liquidity additions. Panel A allocates the increase in stable-facing actions across age bins. Panel B reports the stable-facing share among stable-facing and WETH-facing actions. Panel C classifies transaction origins using their full retained v3 history; transaction origin measures participation, while ownership and project affiliation remain outside the data. Panel D forms a separate 2026 H1 pool cohort for each horizon. Active pools have a retained pool-day during the 15-day window beginning at the horizon, weighted by addition actions on days 0–7. Net flow is vehicle-side additions less removals from day 8 through the horizon, divided by vehicle-side additions on days 0–7. Pool age dates observed pool formation and is distinct from token issuance.

Table 29: Stablecoin circulation, capital growth, and first material links

	Stablecoin core	Stablecoin spoke
Panel A. Next-month log capital growth		
Circulating-supply growth [per 10%]	+0.0083	+0.0041
	(0.0069)	(0.0049)
Holm-adjusted p-value	0.946	1.000
Observations	2,274	5,182
Panel B. First material link in the next month		
Circulating-supply growth [pp per 10%]	−0.0019	−0.0002
	(0.0447)	(0.0003)
Holm-adjusted p-value	1.000	1.000
Observations	9,726	9,097,834

Note: Each coefficient relates a 10% increase in asset-wide circulating supply from month $t-1$ to t to liquidity provision in month $t+1$. Capital-growth models absorb stablecoin–endpoint-pair and endpoint-by-month effects and control for initial capital. Formation models retain endpoint–stablecoin links that never form, absorb stablecoin and endpoint-by-month effects, and control for initial capital. A material link reaches $50,000 across Uniswap v2, SushiSwap v2, and Uniswap v3. Standard errors use CR1 clustering by undirected link and stablecoin-month. Continuous supply growth and capital growth are winsorised at the first and 99th percentiles. Holm-adjusted p-values cover the four core/spoke capital-growth and formation coefficients. The supply measure combines issuance, redemption, transaction demand, and portfolio demand.

H.3 Persistence among pairs with greater entry activity

Table 30 repeats the post-entry stablecoin-share regressions after requiring at least five or ten routes on the entry day. The estimates remain close to one-for-one at both horizons. The main persistence result is also present among entrants with a more precisely measured initial allocation.

Table 30: Post-entry persistence among pairs with greater entry activity

Stablecoin route share among entrants with greater first-day activity	Days 1–30		Days 31–120	
Minimum first-day routes	5	10	5	10
Entry stablecoin share [effect pp per 10 pp]	+9.70*** (0.19)	+9.65*** (0.32)	+9.31*** (0.44)	+9.35*** (0.60)
Observations (retrading pairs)	1,629	671	885	452
Eligible entrants	2,192	775	2,192	775
Entry-date clusters	115	93	100	80
Pair controls and equal-pair weights	Yes			

Note: Each column regresses the stablecoin share in the stated later window on the stablecoin share observed on the entry day. The sample requires complete follow-up through day 120 and subsequent trading in the stated window; the entry day is excluded from every later outcome. All columns control for cohort year, endpoint class, log entry routes, direct-route share, and longer-route share and use equal-pair weights. Coefficients and entry-date-clustered CR1 standard errors in parentheses report the outcome change in pp for a 10 pp higher entry share. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

H.4 Capital thresholds, composition, and later adjustment

Table 31 complements the weekly first-use estimates with a threshold dated before adoption. Panel A follows pairs after full-range stablecoin capital first reaches \$10,000 on both legs. For event e, let B^S_{et} and B^W_{et} denote stablecoin and WETH weak-leg capital, and let S^R_{et} denote stablecoin route share. Panel B estimates

$$\begin{aligned} B^S_{et} &= \max_{v \in \mathcal{S}} \min\{K_{i_e v,t-1}, K_{j_e v,t-1}\}, \qquad & B^W_{et} &= \min\{K_{i_e W,t-1}, K_{j_e W,t-1}\}, \\ Q_{et} &= \frac{B^S_{et}}{B^S_{et} + B^W_{et}}, \qquad & S^R_{et} &= \beta Q_{et} + \alpha_e + \lambda_{m(t)} + \eta_{a_e(t)} + \varepsilon_{et}. \end{aligned} \tag{12}$$

Here $m(t)$ is calendar month and $a_e(t)$ groups elapsed event time into seven-day intervals. A supported stablecoin first carries a route within 30 days for 38.3% of events and within 120 days for 47.0% (panel A). A 10 pp larger stablecoin share of weak-leg capital is associated with +6.90 pp more stablecoin route activity during days 0–29 and +8.35 pp during days 30–119 (panel B). The threshold supplies an interpretable event date; the weekly design in

Table 31: Capital thresholds, stablecoin adoption, and route allocation

Quantity	Estimate	Events
Panel A. Route use after the lagged-capital threshold		
First supported stablecoin route within 30 days	38.3%	1,618
First supported stablecoin route within 120 days	47.0%	1,618
Stablecoin used during days 30–119, among first-month adopters	62.4%	620
Stablecoin route share during days 30–119, same pairs	8.2%	620
Panel B. Prior-day weak-leg capital and route allocation		
10 pp higher stablecoin share of relative capital, days 0–29 [pp]	$+6.90^{***}$ (0.70)	1,495
10 pp higher stablecoin share of relative capital, days 30–119 [pp]	$+8.35^{***}$ (0.43)	1,105

Note: Panel A dates the first time DAI, USDC, or USDT has at least $10,000 of prior-calendar full-range reserve value in Uniswap v2 and SushiSwap v2 on both route legs. Every pair used WETH earlier, and eligibility requires its first stablecoin route to occur on or after the event. Panel B estimates Equation (12) with bridge-event effects, calendar-month effects, seven-day event-age controls, and pair- and date-clustered standard errors. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The capital measure covers full-range positions on the two constant-product venues.

Table 5 also retains pairs with zero measured stablecoin capital.

The main bridge event uses only prior-calendar capital. Table 32 gives the complementary event dated by subsequent persistence: positive prior-calendar capital on both legs during at least 24 of the next 30 days. It reproduces the increase in stablecoin use after support, the relation with continuous relative capital, and the difference in first-month adoption between low-capital and more competitive stablecoin bridges. Because the persistence condition uses later capital, this comparison assesses sensitivity to the event-timing definition.

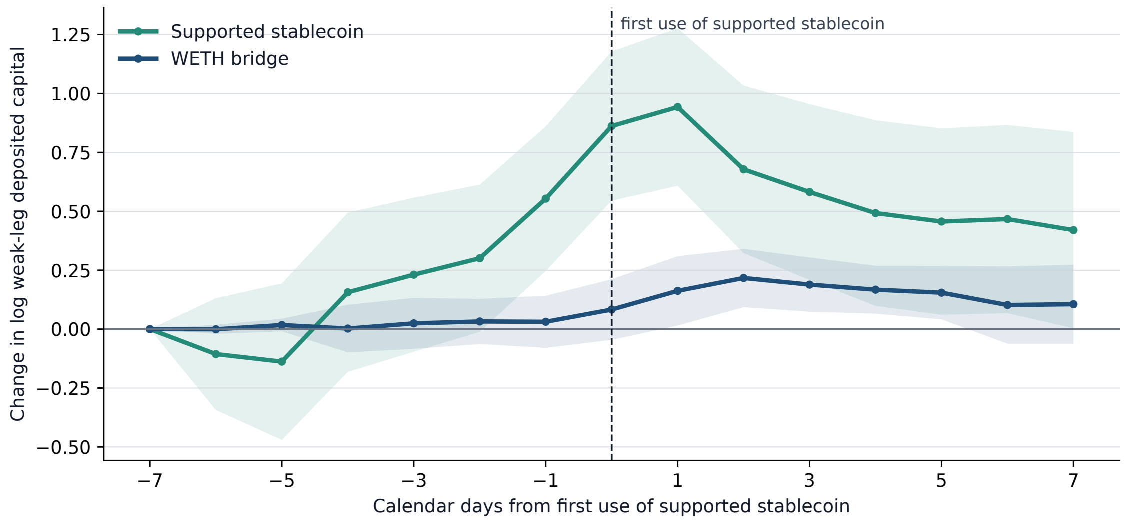
Figure 3 follows events whose supported stablecoin is first used 7 to 119 days after persistent support. Stablecoin weak-leg capital rises during the week before first use, including relative to the matched WETH bridge. The timing is consistent with providers adding capital in anticipation of route demand, traders waiting for sufficient capital on both legs, reserve revaluation, or these forces operating together.

Table 32: Persistent stable-bridge support and route allocation

Outcome	First 30 days	Days 30–119
Panel A. Changes around first persistent stable support		
Stable route share [pp]	$+7.93^{***}$ (1.88)	$+8.68^{***}$ (2.57)
Stable supported-value share [pp]	$+3.11^{**}$ (1.53)	$+6.94$ (4.53)
$\log(1 +$ native routes per active day$)$	-0.20^{***} (0.04)	-0.39^{***} (0.06)
$\log(1 +$ total routes per active day$)$	-0.10^{**} (0.04)	-0.29^{***} (0.05)
$\log(1 +$ native supported value per active day$)$	-0.17 (0.11)	-0.61^{***} (0.19)
$\log(1 +$ total supported value per active day$)$	$+0.06$ (0.10)	-0.32^{*} (0.18)
Bridge events	865	818
Panel B. Stable-bridge competitiveness relative to WETH		
Relative-depth slope [pp per +1 pp]	$+0.64^{***}$ (0.07)	$+0.75^{***}$ (0.05)
Stable route share, depth $< 0.1\times$ WETH [%]	2.4	2.4
Stable route share, depth $\geq$ WETH [%]	53.0 (4.6)	61.2 (13.2)
Stable route share, depth $\geq 2\times$ WETH [%]	69.9 (2.3)	70.7 (9.3)
Active pair-days	11,327	17,027
Panel C. Adoption of supported stablecoin after persistent support		
Outcome	First 30 days	First 120 days
Supported stablecoin used, depth $< 0.1\times$ WETH [%]	42.6	50.5
Supported stablecoin used, depth $\geq 0.1\times$ WETH [%]	84.1	89.7
Difference [pp]	$+41.55^{***}$ (4.95)	$+39.17^{***}$ (4.48)
Difference with controls [pp]	$+25.16^{***}$ (5.76)	$+23.47^{***}$ (5.37)
Difference, no stablecoin endpoint [pp]	$+35.76^{***}$ (11.89)	$+40.26^{***}$ (11.66)

Note: The unit is a pair and its realised single- or cross-venue route class, measured around the first persistent DAI, USDC, or USDT bridge. Support requires positive prior-calendar deposited capital on both legs in Uniswap v2 or SushiSwap v2 on at least 24 of the next 30 calendar days. Every pair has earlier WETH-mediated routes, and the event precedes its first observed stablecoin route. Panel A reports changes from the prior 30 days. Panel B compares the best stablecoin weak-leg capital with WETH weak-leg capital. Panel C records use of a stablecoin in the event's support set within 30 or 120 days. Standard errors are two-way clustered by pair and date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. Deposited capital is an equilibrium pool stock.

Figure 3: Bridge capital around first use of the supported stablecoin



Note: The sample contains 246 pair-by-realised-route-class events whose first use of a stablecoin in the event's support set occurs 7 to 119 calendar days after persistent support. Each line is the equal-event mean change from day -7 in $\log(1 + \text{deposited capital in USD})$ on the weaker leg. Stablecoin capital is the largest bottleneck among the supported stablecoins; WETH capital is the matched incumbent bottleneck. Capital is lagged to the prior calendar day. Shading gives 95% confidence intervals from standard errors two-way clustered by pair and adoption date.

Table 33 separates newly active pools from capital changes in pools already supporting the route. Newly active pools hold 7.5% of stablecoin weak-leg capital on the first-use day; 92.5% lies in pools active one week earlier. Continuing-pool capital rises more on the stablecoin side than on the matched WETH side by +0.16 log point, alongside the smaller pool-activation margin.

H.5 Relative-price risk and deposited bridge capital

Lower endpoint–vehicle relative-price risk is associated with larger constant-product weak-leg capital at the measurement date and 30 days later. This relation is consistent with divergence risk entering liquidity placement. Panel B of Table 34 gives the corresponding vehicle comparison. Stablecoin bridges have higher median risk than WETH bridges and carry the lower relative volatility in fewer than one third of comparable pair-months. The vehicle comparison points to expected fees, hedging, issuer support, and network experience

Table 33: Pool activation and capital growth before stablecoin route use

Outcome	Stablecoin	WETH	Difference
Weak-leg deposited capital [log change]	$+0.76^{***}$	$+0.06$	$+0.70^{***}$
	(0.16)	(0.07)	(0.17)
Active pool count [log change]	$+0.04^{***}$	-0.01^{*}	$+0.05^{***}$
	(0.01)	(0.01)	(0.01)
Any newly active pool [pp]	$+7.32^{***}$	$+0.41$	$+6.91^{***}$
	(1.66)	(0.41)	(1.72)
New-pool share of adoption capital [pp]	$+7.47^{***}$	$+0.41$	$+7.14^{***}$
	(1.70)	(0.41)	(1.77)
Continuing-pool capital [log change]	$+0.23^{***}$	$+0.07^{*}$	$+0.16^{**}$
	(0.07)	(0.04)	(0.08)

Note: The sample contains the bridge events in Figure 3. Stablecoin and WETH outcomes use each vehicle's weaker leg. A newly active pool has positive prior-calendar deposited capital on day zero and none on day -7; a continuing pool has positive capital on both dates. Each cell is an equal-event mean. Parentheses contain standard errors two-way clustered by pair and adoption date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

for the aggregate rotation.

Table 34: Relative-price risk and deposited bridge capital

Log weak-leg capital	(1) Capital at t	(2) Capital at t	(3) Capital at $t+30$	(4) Capital at $t+30$
Panel A: Relative-price risk and full-range capital				
Relative volatility [10 pp]	-0.117^{**} (0.049)		-0.093^{***} (0.029)	
Daily divergence loss [1 bp]		-0.053^{**} (0.021)		-0.044^{**} (0.020)
Log prior vehicle-route count	$+2.218^{***}$ (0.097)	$+2.235^{***}$ (0.095)	$+0.613^{***}$ (0.066)	$+0.625^{***}$ (0.067)
Log initial weak-leg capital			$+0.733^{***}$ (0.022)	$+0.734^{***}$ (0.022)
Within R^2	0.322	0.321	0.702	0.702
Observations	58,447	58,447	58,447	58,447
Pair clusters	4,254	4,254	4,254	4,254
Date clusters	71	71	71	71
Pair × date fixed effects	Yes	Yes	Yes	Yes
Vehicle fixed effects	Yes	Yes	Yes	Yes
Two-way clustered s.e.	Pair, date	Pair, date	Pair, date	Pair, date

Panel B: Native and stablecoin bridge risk	Relative volatility	Daily divergence loss
Native median	126.9%	5.60 bp
Stablecoin median	148.4%	7.68 bp
Stablecoin has lower risk [pair-months]	29.4%	28.6%

Note: The unit is a pair, monthly date, and vehicle among WETH, DAI, USDC, and USDT. Each row requires at least ten routes for the pair during the preceding calendar month, at least 20 daily return observations during days -30 through -1, and positive deposited capital on both route legs for at least one native and one stablecoin bridge in the pair-month. Weak-leg capital is the smaller deposited-capital stock across the two legs in full-range Uniswap v2 and SushiSwap v2 pools, measured at the prior-calendar state. Relative volatility is the larger annualised endpoint–vehicle volatility across the two legs. Daily divergence loss applies the exact 50–50 constant-product expression to the same relative returns and takes the larger leg value. Columns 3 and 4 control for initial weak-leg capital; every column controls for prior vehicle-route activity and absorbs pair-by-date and vehicle effects. Standard errors are two-way clustered by pair and date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The estimates describe equilibrium liquidity allocation; fee income, provider hedges, and provider-level returns are unavailable.

H.6 ETH stress, liquidity supply, and route use

ETH stress affects three distinct objects: external liquidity flows, the dollar capital in a fixed set of pools, and route-specific weak-leg capital. Table 35 relates the first and third objects to executable output and route use. Table 36 decomposes the second into the $\sqrt{k}$ and unit-value components.

Panel A compares stablecoin-facing and WETH-facing pools attached to the same endpoint and week. Higher ETH volatility predicts similar coefficients for v3 additions and withdrawals, while the net-supply coefficient is +0.0003. The price-neutral v2 net-liquidity-unit coefficient is +0.0003. The v3 coefficient on a 0.10-log-point fall in ETH's dollar price is +0.0027 with Holm-adjusted $p = 0.067$; its gross-flow components are individually imprecise. These are differential stable-minus-WETH external-flow responses. Total provider activity and within-pool range management are separate margins.

Panel B holds the pair, trade size, pre-transaction state, and public venue set fixed. A 10 percentage point larger stablecoin share of joint weak-leg dollar capital predicts a +19.10 bp exact-output advantage and +2.87 pp higher probability of stablecoin use (second row). A 100 basis point stablecoin output advantage predicts +10.33 pp higher stablecoin use (third row). In this monthly route-opportunity sample, trailing ETH returns have small coefficients for route-specific weak-leg capital, output, and route choice (first row). A larger stablecoin share of measured V2 and SushiSwap v2 weak-leg capital is associated with better stablecoin quotes and more stablecoin use in this common-support sample.

Table 35: ETH stress, liquidity supply, and route use

Panel A. Stablecoin-minus-WETH liquidity-supply responses

Outcome in week $t+1$	10 pp higher ETH volatility	ETH price fall [0.10 log point]
v3 additions, $\ln(1 + A/K)$	+0.0045 (0.0024)	+0.0041 (0.0065)
v3 withdrawals, $\ln(1 + W/K)$	+0.0044 (0.0023)	+0.0002 (0.0064)
v3 net supply, $\operatorname{asinh}((A - W)/K)$	+0.0003 (0.0004)	+0.0027* (0.0011)
v2 net liquidity units, $\operatorname{asinh}(\Delta L/\sqrt{k})$	+0.0003 (0.0002)	−0.0002 (0.0008)

Panel B. ETH returns, weak-leg capital, quotes, and route use

Regressor	Log stable/WETH USD capital	Stablecoin output lead [bp]	Stablecoin chosen [pp]
ETH return, days −30 to −1 [0.10 log point]	−0.0005 (0.0238)	+0.15 (1.05)	+0.07 (0.25)
Stable share of joint weak-leg USD capital [10 pp]		+19.10*** (2.90)	+2.87*** (0.50)
Stablecoin exact-output advantage [100 bp]			+10.33*** (0.69)
Observations	24,313	24,313	24,313

Note: Panel A compares stablecoin-facing and WETH-facing pools for the same endpoint and week. ETH volatility and returns are measured during week t; liquidity outcomes occur during week $t+1$. Additions and withdrawals are log one plus dollar flow divided by initial pool capital. Net supply is the inverse hyperbolic sine of net dollar flow divided by initial capital. The v2 liquidity-unit outcome divides net liquidity units by initial square-root reserves and is price-neutral. Pools have at least $50,000 of capital. Models absorb endpoint-by-week and pool effects, control for prior fee yield, endpoint–vehicle relative-price volatility, prior additions and withdrawals, capital, and pool age, and cluster standard errors by pool and week. Panel B uses exact two-leg opportunities for which stablecoin and WETH routes are feasible and both prior-calendar weak-leg full-range capital measures are positive. Dollar capital is deposited Uniswap v2 and SushiSwap v2 capital on the weaker leg and records a mark-to-market pool state. Exact output holds the pair, input, pre-transaction state, and public venue set fixed. Panel B models absorb pair and month-of-year effects, control for log input value, ETH volatility, and linear calendar time, and cluster standard errors by pair and exact date. Because ETH return is common within each date, Panel B uses month-of-year effects in place of date effects. Stars in panel A and the first row of panel B use Holm-adjusted p-values within their declared outcome families. The three capital–price–choice links form a separate Holm family. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The estimates are predictive associations.

For each endpoint, date, and exchange in Table 36, both vehicle families have at least \$50,000 in anchor-date pools. We hold those pools fixed and require every one to retain a positive state at the horizon. The resulting sample contains 78,262–79,583 endpoint intervals per horizon; complete follow-up ranges from 90.0–93.5% across venue–horizon cells. A 0.10-log-point fall in ETH's dollar price corresponds to more stablecoin-relative dollar capital at every horizon (column 1). The $\sqrt{k}$ contribution is negative and imprecise after Holm adjustment (column 2), while the three- and seven-day unit-value estimates are economically close to the conditional +0.0500 constant-product reference (column 3). The same WETH price series enters the predictor and WETH-side dollar capital. This stock decomposition is therefore an accounting comparison; $\sqrt{k}$ also includes retained fees and token transfers, while decoded provider additions and withdrawals remain the separate flow outcomes in Table 35, panel A. Measuring cross-price convergence requires transaction-level pool-price deviations and remains outside this table.

Table 36: ETH declines and stablecoin-relative constant-product capital

Horizon	Stablecoin minus WETH [log points per 0.10-log-point ETH price fall]			Obs.
	Total capital, $\Delta \ln V$	Invariant-unit component, $\Delta \ln \sqrt{k}$	Unit value, $\Delta \ln(V/\sqrt{k})$	
1 day	+0.0437*** (0.0091)	−0.0176 (0.0089)	+0.0614 (0.0016)	79,583
3 days	+0.0466*** (0.0057)	−0.0082 (0.0056)	+0.0548 (0.0006)	78,997
7 days	+0.0449*** (0.0039)	−0.0077 (0.0040)	+0.0526 (0.0006)	78,262

Note: Each entry is the coefficient on a 0.10-log-point fall in ETH's dollar price, about 9.5%, in a regression of the indicated stablecoin-minus-WETH log change. The pooled sample combines Uniswap v2 and SushiSwap v2 and contains nonstable endpoints with at least $50,000 in both vehicle families at the anchor. Anchor-date pools are held fixed, and every selected pool must retain a positive validated future state; complete intervals span 90.0–93.5% across venue–horizon cells. Models absorb venue-by-endpoint and anchor-month effects and use anchor-date HAC standard errors with seven lags. Capital obeys $V = \sqrt{k}(V/\sqrt{k})$. Changes in $\sqrt{k}$ include liquidity-provider actions, retained swap fees, and token transfers. Unit value reflects token prices and reserve adjustment. The WETH price series also enters WETH-side dollar capital; 96.7% of primary intervals lack a validated endpoint price, so capital there equals twice the vehicle-side reserve value. Stars on total capital and the invariant-unit component use Holm-adjusted p-values across the 27 venue–horizon–outcome estimates. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The conditional constant-product unit-value reference is +0.0500.

I Additional exact-price and route-choice results

I.1 The first sampled contest after pair entry

The main persistence estimates include pairs for which one vehicle family may remain infeasible. We therefore locate the first sampled monthly date after entry on which the same observed trade can use either WETH or a stablecoin path under the exact common-support rules. Entry requires at least $5,000 of coherent route value on the pair's first active day. Among 118,447 entrants, 580, or 0.5%, reach a sampled exact contest. The median lag is 115.5 days, and 53.3% reach it within 120 days (Table 37, panel A).

For route r on pair p's first sampled date t on which both vehicle families are feasible, we estimate

$$C_{rpt} = \alpha + \beta A^O_{rpt} + \gamma A^K_{pt-1} + \beta_q \log q_r + \delta_t + \eta_i + \xi_j + \phi_s + \varepsilon_{rpt}. \tag{13}$$

In panel B, C_{rpt} indicates stablecoin use or retention of the entry family, and A^O_{rpt} and A^K_{pt-1}

are oriented accordingly. The entry family carries 83.1% of routes at that first common-feasibility date. Exact output predicts retention; the separate prior-capital estimate is imprecise on this early and selective opportunity set.

I.2 Complete established-pair and crossing specifications

Table 38 reports established-pair vehicle choice as a function of exact-output and prior-capital advantages. Table 39 reports route-share movements and challenger-capital estimates around changes in exact-price leadership.

I.3 Execution-cost consequences

Table 40 reports the gross and gas-adjusted output difference between the realised vehicle and its feasible rival. The value-weighted gross shortfall is 7.4 bp across all contestable routes and 3.3 bp among routes retaining a mature incumbent. The associated economic magnitudes are small in the aggregate and larger among small trades and younger pairs.

Table 37: Vehicle choice when both families first become feasible

Panel A. From original pair entry to the first sampled exact contest

Quantity	Primary value	Additional detail
Original pair-entry cohort	118,447 pairs	2,090 entry dates; $5,000 minimum
First sampled exact contest	580 pairs	995 routes
Entry pairs reaching a sampled contest	0.5%	
Entry-to-contest lag [days]	115.5 median	21–318.5 interquartile range
Contest reached within 120 days	53.3%	
Entry family retained, clear comparisons	83.1% of routes	84.4% of pairs
Positive prior-day V2 capital for both families	220 routes	115 pairs

Panel B. Route choice at the first sampled exact contest

Outcome; effects [pp]	(1) Stablecoin chosen	(2) Entry family retained	(3) Entry family retained
Stablecoin current exact-output advantage [per 100 bp]	$+6.39^{***}$ (0.75)		
Entry-family current exact-output advantage [per 100 bp]		$+10.31^{**}$ (3.93)	$+10.20^{**}$ (4.16)
Entry-family prior-day V2 weak-leg capital share [per 10 pp]			+1.85 (4.58)
Log route input value [per log point]	−2.51 (1.61)	−4.01 (4.90)	−3.91 (5.01)
Dependent mean [%]	22.1	84.9	84.9
Within R^2	0.162	0.152	0.152
Observations (routes)	995	219	219
Ordered endpoint pairs	580	114	114
Dates	64	27	27
Prior-day V2 capital positive for both families	No	Yes	Yes
Date, source, destination, and route-scope fixed effects	Yes	Yes	Yes
Pair- and date-clustered s.e.	Yes	Yes	Yes

Note: Original pair entry is the first observed activity with at least $5,000 of coherent route value. The common-feasibility date is the first monthly exact-state date after entry on which both a WETH path and a path through DAI, USDC, or USDT are feasible for an observed route. Panel A describes the resulting lifecycle and retains pairs with a clear native or stablecoin entry family for the retention rows. Panel B estimates Equation (13). Exact output comes from the pre-transaction state and its advantage is capped at 1,000 bp in absolute value. Capital is prior-calendar full-range Uniswap v2 and SushiSwap v2 capital on the weaker route leg; its regressor is the entry family's share of the two families' capital, centred at 50%. Every model includes log input value and date, source, destination, and an indicator for the realised single- or cross-venue route class. Standard errors are clustered by pair and date. Coefficients and standard errors are in percentage points. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The 0.5% coverage rate describes this sampled opportunity set.

Table 38: Exact prices, full-range weak-leg capital, and vehicle retention

Outcome; estimates [pp]	(1) Stablecoin chosen	(2) Incumbent retained	(3) Incumbent retained	(4) Incumbent retained
Stablecoin route has higher exact output	+57.59*** (2.84)			
Challenger route has higher exact output		−58.08*** (2.82)		
Challenger output lead × stable incumbent		+4.47 (7.82)		
Incumbent exact-output advantage [100 bp]			+10.56*** (0.83)	+10.13*** (0.77)
Incumbent lagged full-range capital-share advantage [10 pp]				+2.77*** (0.61)
Log input value [USD]	+0.32 (0.43)	−0.66 (0.46)	−1.56* (0.87)	−1.32* (0.78)
Dependent mean [%]	26.5	73.7	69.7	69.7
Within R^2	0.372	0.382	0.145	0.153
Observations	49,212	34,439	17,778	17,778
Pair clusters	2,596	1,981	678	678
Date clusters	73	73	73	73
Pair fixed effects	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes
Two-way clustered s.e.	Pair, date	Pair, date	Pair, date	Pair, date
Exclusive entry, age $\geq$ 30 days	No	Yes	Yes	Yes
Prior-day full-range capital positive, both vehicles	No	No	Yes	Yes
Minimum absolute output difference [bp]	1	1	1	1
Continuous output gap cap [bp]			1,000	1,000

Note: The unit is a realised two-leg route on the fifteenth day of each month from June 2020 through June 2026. For the same pair, input amount, and pre-transaction state, we quote the best WETH route and the best stablecoin route through DAI, USDC, or USDT across Uniswap v2, SushiSwap v2, and Uniswap v3. Input value is at least \$100, leg values agree within 20%, both vehicle families are feasible, and every observed and quoted leg remains below 5% own-price impact. Ties within one basis point are omitted, and all prices are gross of gas. Column 1 studies stablecoin use. Columns 2 through 4 study retention at least 30 days after exclusive entry. Columns 3 and 4 use the same observations with positive prior-day full-range constant-product weak-leg capital for both families; column 4 estimates Equation (4). Every column includes pair and date fixed effects. Standard errors are clustered by pair and date. Coefficients and standard errors are in percentage points. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 39: Price-rank crossings, route reallocation, and weak-leg capital

Panel A. Incumbent route share around an exact-price rank crossing [pp]

Event month	All crossings	Stable challenger	Native challenger
-3	54.7 (2.0)	73.7 (3.0)	35.4 (3.6)
-2	54.8 (2.4)	73.6 (3.5)	35.7 (4.6)
-1	66.9 (1.8)	83.1 (2.7)	50.5 (3.8)
0 (crossing)	38.1 (1.9)	53.6 (3.8)	22.5 (2.7)
+1	49.2 (2.5)	69.2 (4.3)	29.0 (4.4)
+2	55.4 (2.3)	77.9 (4.2)	32.5 (4.2)
+3	50.4 (2.5)	73.4 (3.6)	27.0 (3.9)

Panel B. Immediate response and next-month price-rank persistence

	Incumbent-share change	Challenger leads next month		Actual vs. placebo
	Activity floor (1)	Activity floor (2)	All crossings (3)	Activity floor (4)
Challenger weak-leg capital share, $Q_{e,-1}$ [10 pp]	$+0.28$ (0.66)	$+3.70^{**}$ (1.52)	$+3.85^{***}$ (0.95)	
Actual crossing [vs. months -3 to -2]				-28.98^{***} (4.71)
Actual crossing × challenger capital [10 pp]				$+2.75$ (1.87)
Crossing controls	Yes	Yes	Yes	Yes
Crossing-month fixed effects	Yes	Yes	Yes	No
Crossing-event fixed effects	No	No	No	Yes
Observations	361	286	583	440
Crossing events	361	286	583	220
Ordered endpoint pairs	163	122	229	83

Note: A price-rank crossing occurs when the pair-month median exact-output gap moves from at least one basis point in favour of the incumbent in month $t-1$ to at least one basis point in favour of the challenger in month t. The crossing sample requires at least two common-support routes and $1,000 of observed input in both crossing months. Panel A gives equal-event means for the 147 crossings observed throughout months -3 to $+3$; parentheses contain pair-clustered standard errors. In panel B, columns 1 through 3 control for challenger family, crossing- and prior-month exact-output gaps, prior incumbent route share, and log crossing-month route activity and include crossing-calendar-month effects. Standard errors are clustered by pair and calendar month. Column 4 stacks the crossing change with the same event's earlier change and includes event effects; standard errors are clustered by pair. Weak-leg capital is the smaller prior-calendar full-range capital stock across the two required legs in Uniswap v2 and SushiSwap v2. Coefficients and standard errors are in percentage points. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 40: Gross and gas-adjusted output shortfalls across feasible vehicle families

Measure	Estimate	Routes
Panel A. All contestable routes		
Observed family has an exact-output shortfall above 1 bp [%]	12.9%	52,477
Conditional median shortfall [bp]	27.2	6,786
Conditional 90th-percentile shortfall [bp]	171.5	6,786
Input-value-weighted mean shortfall [bp]	7.4	52,477
Panel B. Routes retaining a mature exclusive incumbent		
Observed family has an exact-output shortfall above 1 bp [%]	10.7%	27,215
Conditional median shortfall [bp]	17.1	2,901
Input-value-weighted mean shortfall [bp]	3.3	27,215
Panel C. Input-value-weighted mean shortfall by pair age		
Pairs aged 0–89 days [bp]	16.8	13,159
Pairs aged 90–364 days [bp]	19.0	13,166
Pairs aged at least 365 days [bp]	2.1	26,152

Panel D. Receipt-matched routes, gross and gas-adjusted output

	Routes	Lower-output route [%]		Weighted shortfall [bp]	
Input value		Gross	Net of gas	Gross	Net of gas
All routes	52,207	13.0	13.7	7.46	7.65
$100–999	21,371	13.2	14.7	16.02	22.01
$1,000–9,999	22,919	12.8	13.0	9.20	9.51
$10,000–99,999	7,317	13.1	13.1	9.20	9.21
$100,000 and above	600	9.7	9.7	3.78	3.77

Note: The unit is a realised two-leg route in the common-support sample underlying Table 38. A shortfall occurs when the observed vehicle family returns more than one basis point less output than its feasible rival for the same pair, input amount, and pre-transaction state. Conditional quantiles use routes with such a shortfall; input-value-weighted means use every route in the stated sample and assign zero below the one-basis-point threshold. Panel B retains routes at least 30 days after exclusive entry and on which the route keeps the incumbent family. Panel C measures age from the pair's first observed supported date. Panel D uses the same 52,207 routes in its gross and gas-adjusted columns. Path gas is predicted from year, transaction-callee class, ordered venue sequence, leg count, and cross-venue status; the comparison holds observed callee class and effective gas price fixed. Appendix Table 15 reports held-out accuracy and path-specific bounds.

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